Crisis-Dependent Linkages in Major Exchange Rates

Abstract

This paper investigates why some global crises forge long-term links among major currencies, while others fracture them. We address two important questions in currency markets during crises: (1) whether cointegration relationships of major currencies are preserved across distinct crisis episodes, and (2) whether long-horizon currency returns exhibit lower correlation as currency autonomy strengthens. We present a model in which exchange rates load on global stochastic trends under autonomy. The model implies that the survival of cointegrating relations is determined jointly by the rank of the pass-through matrix and the level of autonomy. We then take our model to market data and empirically analyze five major currency pairs against the U.S. dollar (USD): EUR/USD, GBP/USD, JPY/USD, CHF/USD, and AUD/USD across three regimes: the dot-com crash (1997–2000), the Global Financial Crisis (2007–2009), and the COVID-19 pandemic (2020–2022). We find that the Global Financial Crisis (GFC) uniquely produced a cointegrating relation, with deviations corrected by EUR/USD, CHF/USD, and GBP/USD over two- to five-week half-lives, while the dot-com crash and the COVID-19 pandemic exhibit no cointegration. Our long-horizon correlation results further show that weaker currency return comovement during a crisis is associated with weaker or absent cointegration. Our results also demonstrate that crises differ in their long-run effects: only when shocks are both globally systemic and USD-centered are stable long-run anchors preserved. Overall, we show that systemic USD-centered stress compresses autonomy, which causes currencies to load similarly and yields a stable anchor, while heterogeneous or novel persistent shocks dissolve this cointegration.

Keywords: Foreign exchange markets; financial crises; exchange rate cointegration; stochastic trends; crisis regimes; safe-haven currencies.

JEL Classification: F31; C32; G01.

1. Introduction

This paper studies how global crises affect the long-run structure of foreign exchange markets. While a large literature documents heightened volatility, contagion, and short-run comovement during crises, little is known about how systemic stress reshapes the long-run equilibrium relationships among major currencies. In particular, it remains unclear (i) whether currency cointegration relationships are preserved during crises and (ii) how long-horizon currency return comovement evolves as currency autonomy shifts across regimes. Addressing these questions is central to understanding the stability of the international monetary system, the transmission of global shocks, and the limits of diversification in foreign exchange markets. Under standard asset-pricing theory with integrated and frictionless markets, bilateral exchange rates are driven by a common global stochastic discount factor. This typically implies a cointegration rank of $N-1$ in an N-currency system (Backus et al., 2001; Engel and West, 2005). However, structural frictions, heterogeneous monetary regimes, and crisis-specific shocks can alter these long-run linkages. Existing empirical studies largely emphasize short-run volatility spillovers, correlation spikes, or temporary deviations from parity during crises (Flood and Rose, 1995; Melvin and Taylor, 2009; Aloui et al., 2014), while long-run analyses typically assume a stable number of stochastic trends. As a result, the possibility that crises may fundamentally change the long-run stochastic structure of exchange rates remains largely unexplored.

This paper contributes to the literature by providing a unified, regime-dependent analysis of long-run currency linkages across three regimes: the dot-com crash, the Global Financial Crisis (GFC), and the COVID-19 pandemic. We combine a factor-based asset-pricing framework with Johansen cointegration analysis to allow the degree of currency autonomy and the number of persistent global stochastic trends to vary across regimes. Our approach explicitly distinguishes between the existence of long-run equilibrium relationships, captured by cointegration, and long-horizon return comovement, which reflects how tightly currencies load on global risk factors over extended horizons even in the absence of cointegration. Using daily spot exchange rates for major USD currency pairs, we estimate crisis-specific vector error correction models and examine how cointegration rank, adjustment dynamics, and long-horizon correlations evolve across regimes. The results reveal sharp contrasts across crises. During the Global Financial Crisis, currencies exhibit strong cointegration and high long-run comovement, consistent with compressed currency autonomy driven by USD funding stress. In contrast, both the dot-com crash and the COVID-19 pandemic display a breakdown of cointegration. Most notably, during the COVID-19 pandemic, we detect the emergence of an additional stochastic trend, increasing the number of independent persistent shocks and weakening long-run cointegrating relations.

These findings challenge the prevailing crisis-contagion narrative, which implicitly assumes that crises uniformly strengthen long-run currency integration, see (Arshanapalli and Doukas, 1993; Hung and Cheung, 1995; Yang et al., 2003). Instead, our results show that crises can reshape the underlying factor structure of foreign exchange markets in qualitatively different ways, with important implications for asset pricing, risk management, and international portfolio diversification. By linking regime-dependent factor exposure to changes in cointegration rank and long-run comovement, the paper provides a structural interpretation of why some crises reinforce global currency linkages while others fragment them. The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 presents the model setup and develops the theoretical framework. Section 4 describes the data and empirical methodology. Section 5 presents the empirical results and discussion of findings. Section 6 concludes and discusses avenues for future research.

2. Literature review

This paper sits at the intersection of the foreign exchange (FX) literature and asset pricing. We contribute by linking global risk factors and intermediary constraints to regime-dependent changes in long-run currency linkages. A foundational strand of FX research studies common stochastic trends among major exchange rates using cointegration methods. Early work shows that several dollar exchange rates share persistent global drivers, motivating vector error correction models and Johansen rank tests to identify long-run cointegrating relations (Baillie and Bollerslev, 1989; Johansen, 1988, 1991). Asset-pricing perspectives further reconcile the near-random-walk behavior of exchange rates at short horizons with the present-value relevance of fundamentals (Engel and West, 2005). Together, these studies implicitly assume stable common global stochastic trends, typically consistent with $N-1$ cointegrating relations in integrated FX markets. However, they do not address how major global shocks may alter the number of nonstationary factors driving currencies—a central focus of our analysis. Modern currency asset pricing links exchange rates to priced risk factors and intermediary constraints. Cross-sectional currency returns load on common factors such as the dollar and carry, which explain systematic risk premia across currencies (Lustig et al., 2011). Intermediary asset-pricing models show that exchange rates also respond to dealer balance-sheet constraints and shifts in global liquidity, as funding frictions affect currency pricing and international capital flows (Gabaix and Maggiori, 2015). These frameworks provide structural channels—through risk prices and funding conditions—by which global shocks can synchronize currencies while still allowing for cross-country heterogeneity and time variation.

Crisis episodes illustrate how dollar funding conditions transmit globally through FX markets. During the Global Financial Crisis, disruptions in USD funding markets generated widespread dollar shortages outside U.S. banks (Baba and Packer, 2009; McGuire and Von Peter, 2012). Subsequent research interprets these shortages as linked to dealer leverage, regulatory constraints, and balance-sheet costs (Du et al., 2018). These mechanisms imply that under stress, currencies may load more strongly and more uniformly on global funding factors, potentially compressing monetary and financial autonomy even amid turbulent short-run dynamics. The COVID-19 pandemic triggered severe stress in FX markets and unprecedented policy interventions, including expanded central bank liquidity facilities (Schrimpf et al., 2020). Related work shows that these facilities mitigated funding shortages (Bahaj and Reis, 2020). While this literature emphasizes short-run funding stress, it raises an unresolved question: did the COVID-19 pandemic also alter long-run currency linkages in ways that differ from the GFC? From a time-series perspective, changes in the number of stochastic trends correspond to changes in cointegration rank. Structural-break tests in cointegrating relationships permit endogenous shifts in long-run relations (Gregory and Hansen, 1996), while multiple-break frameworks allow inference in higher-dimensional systems (Qu and Perron, 2007). These tools provide a natural empirical framework to assess whether crisis episodes weaken, eliminate, or reconfigure long-run FX integration. Complementary approaches based on variance-decomposition connectedness (Diebold and Yilmaz, 2012) and network methods measure the intensity of comovement, but they do not distinguish between tighter covariance and changes in the number of underlying nonstationary factors.

Our contribution to the literature is fourfold. First, we introduce a no-arbitrage, log-normal stochastic discount factor with global risk factors into a pass-through structure that maps directly into cointegration settings. To the best of our knowledge, this provides the first structural asset-pricing framework in which cointegration rank becomes an endogenous outcome of global financial conditions. Second, we incorporate currency- and regime-specific autonomy parameters governing factor pass-through. These parameters provide an economic interpretation of cross-country heterogeneity in exposure to global shocks, connecting cointegration dynamics to intermediary-based models of international financial resilience (Aizenman and Jinjarak, 2014). Third, we distinguish between the number of common stochastic trends (cointegration rank) and the tightness of long-run comovement (covariance and connectedness). This distinction clarifies how crises can simultaneously produce high correlation yet lower cointegration rank if additional nonstationary global factors emerge. Fourth, we show that while the GFC is associated with strengthened long-run FX integration and compressed autonomy, the COVID-19 pandemic coincides with weaker cointegration among major currencies. This pattern is consistent with the emergence of new global stochastic trends, altered funding mechanisms, and novel policy backstops such as central bank swap lines, updating the crisis-integration paradigm in light of the modern dollar funding system (Schrimpf et al., 2020). Together, our paper bridges classic FX cointegration analysis with contemporary finance theory, offering an explanation for why long-run currency linkages can strengthen during one crisis yet weaken in another.

3. Model setup

We now present a regime-dependent factor model linking the long-run dynamics of N non-USD exchange rates to a set of nonstationary global factors. The U.S. is the numeraire for exposition. Let the spot exchange rate be $S_{i,t}$ = units of currency i per USD $\quad (i = 1, \ldots, N)$ so an increase in $S_{i,t}$ indicates an appreciation in USD against the currency i. Define $X_{i,t} = \log S_{i,t}$, $\boldsymbol{X}_t = (X_{1,t}, \ldots, X_{N,t})'$. For notational convenience, the model is written using the foreign currency per USD convention, i.e., USD/currency$_i$, to analyze the representative currency pair, so that $S_{i,t}$ is the number of units of currency i per USD. In the empirical section, where we analyze and compare multi-currency pairs, we re-express all exchange rates in the standard currency$_i$/USD format (USD per unit of currency i) to harmonize the data across currencies and report results in dollar terms. Since $\log(S_{i,t}^{-1}) = -X_{i,t}$, switching conventions only changes the sign of $X_{i,t}$ and does not affect any analysis or inference in this paper. The rationale behind constructing the proposed model is grounded in an empirical observation and in salient crisis-era developments: crisis episodes differ not only in intensity but also in the number and nature of persistent global shocks affecting major currency markets. Standard exchange rate models typically assume a fixed long-run structure—meaning a constant number of stochastic trends and stable factor loadings across time. However, recent evidence from global liquidity, dollar funding stress, and macro-financial fragmentation during crises suggests that systemic disruptions can introduce new persistent drivers (e.g., dollar-funding shocks in the GFC or heterogeneous macro shocks during the COVID-19 pandemic). Our model is therefore constructed to allow the long-run factor structure to change across crisis regimes by permitting the loading matrix and the dimension of global stochastic trends to vary. This design also directly matches the empirical objective of the paper: to test whether long-run comovement and cointegration among major USD currency pairs themselves shift across crises. The model provides a coherent theoretical justification for the regime-dependent patterns explored in the data.

3.1. Pricing kernel stochastic discount factor and global factors

We price assets in USD using a log-normal stochastic discount factor (SDF):

$$\log M_{t+1} = -r_t - \tfrac{1}{2}\,\varsigma_r^2 - \xi_{t+1}, \tag{1}$$

where r_t is the USD short rate, ς_r^2 is the total SDF innovation variance, ξ_{t+1} is the stochastic discount factor innovation, and $\mathbb{E}_t[M_{t+1}] = e^{-r_t}$ by construction. In regime r, the innovation admits q_r nonstationary global factors:

$$\xi_{t+1} = \boldsymbol{\lambda}_r^{\top} \Delta \boldsymbol{g}_{t+1} + \upsilon_{t+1}, \quad \mathbb{E}_t[\upsilon_{t+1} \Delta \boldsymbol{g}_{t+1}] = 0. \tag{2}$$

Here, $\boldsymbol{g}_t = (g_{1,t}, \ldots, g_{q_r,t})'$ denotes the q_r independent $I(1)$ global factors in regime r, which may capture persistent forces such as USD funding stress, global risk sentiment, or commodity-price shocks; $\boldsymbol{\lambda}_r \in \mathbb{R}^{q_r}$ represents the corresponding stochastic discount factor loadings in regime r; and $\upsilon_{t+1} \sim \mathcal{N}(0, \sigma_\upsilon^2)$ is an idiosyncratic USD pricing-kernel shock that is orthogonal to $\boldsymbol{g}_{t+1}$. Each factor follows a random walk with regime-dependent drift and volatility:

$$g_{k,t+1} = g_{k,t} + \mu_{k,r} + \sigma_{g_k,r}\,\varepsilon_{k,t+1}, \quad \varepsilon_{k,t+1} \overset{i.i.d.}{\sim} \mathcal{N}(0,1), \quad k = 1, \dots, q_r. \tag{3}$$

The total stochastic discount factor innovation variance is:

$$\varsigma_r^2 = \boldsymbol{\lambda}_r'\,\Sigma_{g,r}\,\boldsymbol{\lambda}_r + \sigma_\upsilon^2, \quad \Sigma_{g,r} = \operatorname{diag}(\sigma_{g_1,r}^2, \dots, \sigma_{g_{q_r},r}^2).$$

Currencies can differ in their exposures to the vector of global factors (e.g., global risk, liquidity conditions, macroeconomic policy shifts), which means that the common-factor block loads differently on each currency: for some currencies, one or more factor loadings are large (high coefficients along specific factor directions), while for others they are small or negligible. In (2), the innovation to the stochastic discount factor is decomposed into two conceptually distinct components: a *systematic* term, $\boldsymbol{\lambda}_r^\top \mathbf{g}_{t+1}$, and an *idiosyncratic* term, υ_{t+1}. The q_r-dimensional vector $\mathbf{g}_{t+1}$ captures shocks that are common to all currencies, though not necessarily equally important; examples include shifts in USD funding conditions, coordinated changes in major central bank policy stances, or global risk-on/risk-off episodes. The vector $\boldsymbol{\lambda}_r$ measures the sensitivity of the *pricing kernel* to each of these common shocks in regime r, and in the affine solution each currency i inherits its own q_r-vector of pass-through coefficients $\boldsymbol{\lambda}_{i,r}$, reflecting its specific exposure profile to the different global factors. By contrast, υ_{t+1} represents purely currency-specific noise in the stochastic discount factor, orthogonal to $\mathbf{g}_{t+1}$ by construction. Its variance σ_υ^2 fully determines its scale, so no additional factor loadings are needed at the stochastic discount factor level. Economically, the vector $\boldsymbol{\lambda}_r$ captures how each orthogonal component of the common-factor block propagates through the pricing kernel, while the heterogeneous $\boldsymbol{\lambda}_{i,r}$ map those common shocks into individual currency returns; the idiosyncratic term υ_{t+1} is already asset-specific and its effects are entirely absorbed by the currency it directly influences.

3.2. Payoffs, no-arbitrage, and an affine solution

A one-period claim pays $D_{i,t+1}$ units of currency i; its USD price is

$$P_{i,t} = \mathbb{E}_t\left[M_{t+1}\frac{D_{i,t+1}}{S_{i,t+1}}\right]. \tag{4}$$

For a pure foreign exchange claim delivering one unit of currency i (i.e., $D_{i,t+1} = 1$), replication implies $P_{i,t} = 1/S_{i,t}$. No-arbitrage gives:

$$\frac{1}{S_{i,t}} = \mathbb{E}_t\left[M_{t+1}\frac{1}{S_{i,t+1}}\right],$$

or equivalently,

$$1 = \mathbb{E}_t\left[M_{t+1}\frac{S_{i,t}}{S_{i,t+1}}\right] = \mathbb{E}_t\left[M_{t+1}e^{X_{i,t}-X_{i,t+1}}\right], \tag{5}$$

our key foreign exchange pricing restriction. The intuition behind the no-arbitrage condition is that the dollar price paid today for a future claim of one unit of foreign currency

$$\mathbb{E}_t\left[M_{t+1}\frac{1}{S_{i,t+1}}\right],$$

must equal the current dollar cost of acquiring that same unit of foreign currency:

$$\frac{1}{S_{i,t}}.$$

If this equality failed, arbitrageurs could construct riskless profit by trading currencies and bonds, contradicting the absence of arbitrage in equilibrium. Risk-bearing profits remain possible but are not part of this no-arbitrage condition.

3.3. Solving for exchange-rate levels

Assuming (i) the stochastic discount factor is logarithmically normal as in (1)-(2) and (ii) the fundamentals $\boldsymbol{z}_t$ $(K \times 1)$ are stationary and independent of global factor innovations, the log-linear solution takes the affine form

$$X_{i,t} = \mu_{i,r} + \boldsymbol{\lambda}_{i,r}'\,\boldsymbol{g}_t + \Gamma_i'\boldsymbol{z}_t + u_{i,t}, \tag{6}$$

with $u_{i,t}$ stationary, and

$$\mu_{i,r} := -\bar{r}_r - \tfrac{1}{2}\varsigma_r^2 - (\lambda_{i,r} + \lambda_r)^\top \mu_r + \kappa_{i,r}, \tag{7}$$

$$\kappa_{i,r} \equiv \tfrac{1}{2}\Big[(\lambda_{i,r} + \lambda_r)^\top \Sigma_{g,r}(\lambda_{i,r} + \lambda_r) + \sigma_\upsilon^2 + \mathrm{Var}_t\big(\Delta(\Gamma_i^\top z_t + u_{i,t})\big)\Big], \tag{8}$$

$$\Gamma_i'\boldsymbol{z}_t = \sum_{k=1}^{K}\gamma_{i,k}\,z_{k,t}, \quad \boldsymbol{z}_t = (z_{1,t},\ldots,z_{K,t})' \text{ is stationary,} \tag{9}$$

$$u_{i,t} = X_{i,t} - \mu_{i,r} - \boldsymbol{\lambda}_{i,r}'\boldsymbol{g}_t - \Gamma_i'\boldsymbol{z}_t, \quad u_{i,t} \text{ stationary.} \tag{10}$$

Here $\bar{r}_r \equiv \mathbb{E}[r_t \mid r]$ is the regime-r mean short rate; $(\boldsymbol{\mu}_r, \boldsymbol{\Sigma}_{g,r}, \boldsymbol{\lambda}_r)$ are regime parameters of the q_r-dimensional global-factor process and stochastic discount factor; and σ_υ^2 is the variance of the stochastic discount factor idiosyncratic residual. The global-factor vector $\boldsymbol{g}_t$ enters linearly in the pricing kernel, so $X_{i,t}$ must load linearly on $\boldsymbol{g}_t$. Currencies differ in their exposure profiles to the q_r orthogonal global factors, with some exhibiting large loadings in certain factor directions and others having smaller or near-zero loadings. This heterogeneity is captured by the currency-specific vector $\boldsymbol{\lambda}_{i,r}$, while $\boldsymbol{\lambda}_r$ measures the sensitivity of the pricing kernel to each global factor in regime r. In the stochastic discount factor, the innovation decomposition $\xi_{t+1} = \boldsymbol{\lambda}_r^\top \Delta\boldsymbol{g}_{t+1} + \upsilon_{t+1}$ separates the *systematic* component, $\boldsymbol{\lambda}_r^\top \Delta\boldsymbol{g}_{t+1}$, from the *idiosyncratic* component, υ_{t+1}. The vector $\Delta\boldsymbol{g}_{t+1}$ represents shocks common to all currencies—such as shifts in global risk appetite, changes in major reserve currency funding conditions, or synchronized monetary policy moves—but whose economic impact varies across currencies according to their $\boldsymbol{\lambda}_{i,r}$ profiles. By contrast, υ_{t+1} is orthogonal to $\Delta\boldsymbol{g}_{t+1}$ by construction, representing purely currency-specific noise in the stochastic discount factor; its variance σ_υ^2 fully determines its scale, so no additional factor loadings are needed for this term at the SDF level. We note that the use of $\Delta\boldsymbol{g}_{t+1}$, rather than $\boldsymbol{g}_{t+1}$, ensures that SDF innovations are stationary. Since $\boldsymbol{g}_t$ is assumed to be integrated of order one with stationary increments, modeling shocks in terms of $\Delta\boldsymbol{g}_{t+1}$ prevents the SDF from inheriting nonstationarity and preserves internal consistency with the cointegration structure. The vector $\boldsymbol{z}_t$ is taken to be stationary so that exchange rates can load on *mean-reverting* conditions without introducing additional stochastic trends. While short-horizon exchange rates often approximate random walks—weakening their level relations with trending fundamentals[1]—stationary variables capture time-varying but transitory forces that help explain cross-sectional heterogeneity and conditional dynamics.[2] Treating $\boldsymbol{z}_t$ as stationary preserves the common integrated of order one trends driven solely by $\boldsymbol{g}_t$ and leaves the cointegration rank determined by the rank of $\boldsymbol{\Lambda}_r$; standard vector error correction models or cointegrated vector autoregression frameworks allow such stationary exogenous controls without affecting rank tests.[3] Finally, the residual $u_{i,t}$ measures the deviation of $X_{i,t}$ from the level implied by its long-run relation with $\boldsymbol{g}_t$ and $\boldsymbol{z}_t$. If $u_{i,t}$ were nonstationary, it would imply an additional persistent factor orthogonal to $\boldsymbol{g}_t$, contradicting the assumption that $\boldsymbol{g}_t$ spans all nonstationary variation. Stationarity of $u_{i,t}$ ensures that any cointegration between $X_{i,t}$ and $\boldsymbol{g}_t$ is not impaired, and confines $u_{i,t}$ to mean-reverting, thereby preserving the internal consistency of the model's long-run factor structure. The USD price of one unit of currency i delivered at $t+1$ is $1/S_{i,t}$. No-arbitrage implies:

$$1 = \mathbb{E}_t \left[M_{t+1} e^{X_{i,t} - X_{i,t+1}} \right].$$

Solving under log-normality and stationary fundamentals $\boldsymbol{z}_t$ yields:

$$X_{i,t} = \mu_{i,r} + \boldsymbol{\lambda}_{i,r}' \, \boldsymbol{g}_t + \Gamma_i' \boldsymbol{z}_t + u_{i,t}, \tag{11}$$

where $\boldsymbol{\lambda}_{i,r} \in \mathbb{R}^{q_r}$ reflects currency-specific exposure to the q_r global factors, $\boldsymbol{g}_t \in \mathbb{R}^{q_r}$. We now present this result more formally in the lemma that follows.

Lemma 1. *(Affine Representation of Exchange Rates) Suppose the log exchange rate* $X_{i,t} = \log S_{i,t}$ *satisfies the difference equation*

$$\Delta X_{i,t+1} = a_{i,r} + \lambda_{i,r}' \, \Delta g_{t+1} + \Gamma_i' \, \Delta z_{t+1} + \epsilon_{i,t+1}, \tag{12}$$

together with the no-arbitrage restriction

$$\mathbb{E}_t \Big[M_{t+1} e^{X_{i,t} - X_{i,t+1}} \Big] = 1.$$

Assume the following:

A1 *Log-normal Stochastic Discount Factor: The stochastic discount factor satisfies*

$$\log M_{t+1} = -r_t - \tfrac{1}{2}\varsigma_r^2 - \xi_{t+1}, \quad \xi_{t+1} \sim \mathcal{N}(0, \varsigma_r^2),$$

with $\mathbb{E}_t[M_{t+1}] = e^{-r_t}$.

[1]See Engel and West (2005) for the near–random-walk implication; classic out-of-sample evidence includes Meese and Rogoff (1983).

[2]Taylor-rule fundamentals and related stationary deviations have documented predictive content at certain horizons; see, e.g., Molodtsova and Papell (2009).

[3]See Johansen (1988) vector error correction model framework as commonly implemented.

A2 *Stochastic Discount Factor Shocks: The innovation is driven by global factor shocks,*

$$\xi_{t+1} = \lambda_r' \Delta g_{t+1} + \upsilon_{t+1},$$

with υ_{t+1} stationary and orthogonal to Δg_{t+1}.

A3 *Integration Orders: g_t is integrated of order one with stationary increments Δg_{t+1}, while z_t is integrated of order zero.*

A4 *Orthogonality: $\{\Delta g_{t+1}, \Delta z_{t+1}, \epsilon_{i,t+1}, \upsilon_{t+1}\}$ are mean-zero, jointly Gaussian, with*

$$\mathbb{E}_t[\Delta z_{t+1}\, \Delta g_{t+1}'] = 0, \quad \mathbb{E}_t[\epsilon_{i,t+1}\, \Delta g_{t+1}'] = 0, \quad \mathbb{E}_t[\epsilon_{i,t+1}\, \Delta z_{t+1}'] = 0.$$

Then the log exchange rate admits the affine representation

$$X_{i,t} = \mu_{i,r} + \lambda_{i,r}'\, g_t + \Gamma_i' z_t + u_{i,t},$$

where $u_{i,t}$ is integrated of order zero and $\mu_{i,r}$ absorbs deterministic terms including the risk-adjusted drift identified from the no-arbitrage restriction. Note that a formal proof is provided in the Appendix.

3.4. Regime-dependent pass-through (autonomy)

To model how the sensitivity of exchange rates to the global factor vector changes across regimes, we parameterize the factor loadings for currency i in regime r as

$$\boldsymbol{\lambda}_{i,r} = (1 - \boldsymbol{\psi}_{i,r}) \odot \bar{\boldsymbol{\lambda}}_i, \qquad \bar{\boldsymbol{\lambda}}_i \neq \mathbf{0} \text{ fixed}, \tag{13}$$

where $\boldsymbol{\psi}_{i,r} = (\psi_{i1,r}, \ldots, \psi_{iq_r,r})' \in [0,1]^{q_r}$ is the vector of "autonomy" parameters of currency i from each global factor in regime r, $\bar{\boldsymbol{\lambda}}_i \in \mathbb{R}^{q_r}$ is the baseline (regime-invariant) vector of factor loadings, and $\odot$ denotes the Hadamard (elementwise) product. A lower $\psi_{ik,r}$ implies stronger pass-through from factor k (larger $|\lambda_{ik,r}|$), meaning that the currency's value is more sensitive to movements in that global factor. A higher $\psi_{ik,r}$ implies greater autonomy from factor k in that regime. The following assumptions describe how volatility and autonomy evolve across regimes, and ensure the tractability of the model:

B1 **Crisis volatilities:** $\sigma^2_{g_k,\text{crisis}} \geq \sigma^2_{g_k,\text{noncrisis}}$ for all k, with strict inequality for at least one k. *Crisis periods exhibit heightened volatility in at least one global factor, reflecting increased uncertainty and tighter global financial conditions.*

B2 **Autonomy compression:** $\psi_{ik,\text{crisis}} \leq \psi_{ik,\text{noncrisis}}$ for all i, k. *Market stress and funding pressures force more synchronized exchange-rate responses to the global factors.*

B3 **Stationarity and orthogonality:** $\boldsymbol{z}_t$ and $\boldsymbol{u}_t$ are stationary and orthogonal to factor innovations $\boldsymbol{\varepsilon}_{t+1}$. *This preserves the cointegration rank implied by $\boldsymbol{\Lambda}_r$ and maintains orthogonality between predictable components and shocks.*

For the factor-loading structure for each regime r, we define the $N \times q_r$ pass-through matrix

$$\boldsymbol{\Lambda}_r = \begin{bmatrix} \boldsymbol{\lambda}_{1,r}' \\ \vdots \\ \boldsymbol{\lambda}_{N,r}' \end{bmatrix} = \begin{bmatrix} (1-\psi_{11,r})\,\bar{\lambda}_{11} & \ldots & (1-\psi_{1q_r,r})\,\bar{\lambda}_{1q_r} \\ \vdots & \ddots & \vdots \\ (1-\psi_{N1,r})\,\bar{\lambda}_{N1} & \ldots & (1-\psi_{Nq_r,r})\,\bar{\lambda}_{Nq_r} \end{bmatrix},$$

which stacks the currency-specific loading vectors $\boldsymbol{\lambda}'_{i,r}$ across i. Here, $\boldsymbol{\lambda}_{i,r} \in \mathbb{R}^{q_r}$ denotes the currency-specific loading vector on the q_r global factors in regime r, $\psi_{ik,r}$ represents the autonomy of currency i from factor k in regime r (with lower values implying stronger pass-through from that factor), Γ_i captures the loadings on stationary fundamentals $\boldsymbol{z}_t$ that do not affect the cointegration rank, and $u_{i,t}$ is a stationary mean-reverting residual term. With these definitions, the cross-section of exchange rates in regime r is

$$\boldsymbol{X}_t = \boldsymbol{\mu}_r + \boldsymbol{\Lambda}_r \boldsymbol{g}_t + \Gamma \boldsymbol{z}_t + \boldsymbol{u}_t, \tag{14}$$

where $\boldsymbol{\mu}_r \in \mathbb{R}^N$ collects the regime-specific intercepts, $\boldsymbol{g}_t \in \mathbb{R}^{q_r}$ is the vector of nonstationary global factors, Γ is the $N \times K$ matrix of stationary-factor loadings, and $\boldsymbol{u}_t$ is the vector of stationary residuals.

3.5. Cointegration rank and regime dependence

Given the affine representation, $\boldsymbol{X}_t = \boldsymbol{\mu}_r + \boldsymbol{\Lambda}_r\boldsymbol{g}_t + \Gamma\boldsymbol{z}_t + \boldsymbol{u}_t$, the long-run dimension of comovement among the N exchange rates in regime r is determined by the rank of the pass-through matrix $\boldsymbol{\Lambda}_r$. The vector $\boldsymbol{g}_t \in \mathbb{R}^{q_r}$ collects the q_r nonstationary global factors relevant in regime r, while $\boldsymbol{\Lambda}_r \in \mathbb{R}^{N\times q_r}$ stacks the currency-specific loading vectors $\boldsymbol{\lambda}'_{i,r}$. Stationary fundamentals $\boldsymbol{z}_t$ and residuals $\boldsymbol{u}_t$ do not affect the cointegration rank because they are integrated of order zero. The number of distinct common stochastic trends transmitted to $\boldsymbol{X}_t$ in regime r is $\mathrm{rank}(\boldsymbol{\Lambda}_r) \leq \min(N, q_r)$ This yields the standard cointegration rank result. If $\mathrm{rank}(\boldsymbol{\Lambda}_r) = 1$, there is a single nonstationary common component and the system has $N-1$ cointegrating relations (the single-factor baseline). If $\mathrm{rank}(\boldsymbol{\Lambda}_r) = k > 1$, there are $N-k$ cointegrating relations, meaning multiple independent global trends drive the system. If $\mathrm{rank}(\boldsymbol{\Lambda}_r) = N$, there is no cointegration, since the number of common stochastic trends equals the number of currency variables, so no linear combination can eliminate all trends. If $\mathrm{rank}(\boldsymbol{\Lambda}_r) = 0$, then there is no common stochastic trend: either all trends are idiosyncratic (so stochastic trends cannot be eliminated via linear combinations, implying no cointegration), or there are no stochastic trends at all (so the variables are stationary and therefore not cointegrated). Allowing for regime variation, we note that both the factor volatilities $\sigma^2_{g_k,r}$ and the autonomy parameters $\psi_{ik,r}$ can vary across regimes, altering the effective rank of $\boldsymbol{\Lambda}_r$. For example, a crisis may increase similarity in response to global factors (making common trends more dominant) and/or *compress* autonomy parameters (aligning currencies more closely with the factors). On the other hand, if a new factor emerges—for example, a pandemic-specific risk driver during the COVID-19 pandemic—it can reduce cointegration relations by raising autonomy or increasing the number of stochastic trends. In the extreme, if enough independent trends are introduced, cointegration may disappear even if it was previously present.

Insert Figure 1

To build intuition for the transmission mechanism implied by the model, Figure 1 presents a schematic representation of the exchange rate evolution.

As shown in Figure 1, a regime-dependent set of global stochastic trends, denoted by q_r, drives exchange rate dynamics. These persistent factors are transmitted through currency-specific autonomy filters, $\psi_{i,r}$, which govern the degree of pass-through of common global shocks into exchange rates. When autonomy is low ($\psi \approx 0$), global shocks pass through strongly, resulting in common-factor-dominated exchange rate movements and stronger cointegration across currencies. When autonomy is high ($\psi \approx 1$), pass-through is limited, and exchange rates are driven primarily by idiosyncratic shocks, leading to the weakest or a complete absence of cointegration. Intermediate values ($0 < \psi < 1$) generate mixed dynamics that reflect both common and idiosyncratic components. This structure motivates the formal results that follow, which characterize how variation in q_r and $\psi_{i,r}$ jointly determines cointegration and the effective rank of the pass-through matrix that uncovers the existence of cointegrating structure across currencies.

3.6. Main propositions

The following propositions summarize the key theoretical implications of the multi-factor, regime-dependent pass-through model developed above. They connect the statistical properties of the pass-through matrix $\mathbf{\Lambda}_r$ to observable patterns of cointegration, long-run comovement, and regime shifts in exchange-rate dynamics. Formal proofs are provided in the Appendix.

Proposition 1. *Given regime r and assumptions B1–B3, the cointegration rank of X_t is*

$$\textit{Cointegration rank} = N - \operatorname{rank}(\Lambda_r).$$

This result formalizes the direct link between the number of independent nonstationary global trends and the existence of long-run exchange-rate linkages. If $\operatorname{rank}(\mathbf{\Lambda}_r) = 1$, then all persistent variation in each exchange rate is driven by a single common global factor (e.g., the USD funding/liquidity factor), yielding $N - 1$ cointegrating relationships, meaning the exchange rates have established long-run relationships and are related in the long run in $N - 1$ different ways, where in each case the residual is stationary, with the residual being the spreads between the exchange rate normalized on and a linear combination of the other exchange rates. As $\operatorname{rank}(\mathbf{\Lambda}_r)$ increases, additional nonstationary drivers (for example, stochastic factors like commodity-price trends, regional funding shocks, or segmented risk premia) introduce new sources of long-run divergence, reducing the number of cointegrating relations. In the extreme case $\operatorname{rank}(\mathbf{\Lambda}_r) = N$, each currency contains an independent stochastic trend, so no stationary linear combination exists and cointegration disappears. If $\operatorname{rank}(\mathbf{\Lambda}_r)$ remains finite and bounded during crisis, such that (i) $\sigma^2_{g_k,r}$ rises for one or more factors, and (ii) $\lambda_{ik,r}$ is non-negligible and similar across currencies for each global factor, i.e., react similarly to each global factor and the response is non-trivial, which means there is non-zero convergence in response to the global factors across currencies (autonomy compression, each currency has less autonomy from global factors, that is, the common stochastic trends), then long-run correlations, that is the correlation of long-run currency returns across currencies, increases, and factor volatility explains a larger share of the total variance of the long-horizon returns for each currency.

Proposition 2. *For currencies $i = 1, \dots, N$ and common stochastic trends $k = 1, \dots, q_r,\ q_r \leq N$,*

$$\begin{aligned} &\lambda_{1k} \approx \lambda_{2k} \approx \cdots \approx \lambda_{Nk}, \\ \forall\, i \neq j,\ \forall\, k \in \{1, \dots, q_r\}: \quad &\lambda_{ik} \approx \lambda_{jk}, \quad \lambda_{ik} \neq 0,\ \lambda_{jk} \neq 0. \end{aligned}$$

then the correlation of long-horizon returns between currencies i and j increases toward one:

$$\rho_{ij}(H) \equiv \mathrm{Corr}\big(r_{i,H}, r_{j,H}\big) \ \to \ 1,$$

where $r_{i,H}$ denotes the H-period return of currency i.

When the long-horizon correlation among currency returns is low and moves away from one, this suggests heterogeneous responses to the common stochastic trends (currencies have greater autonomy from the common stochastic trends). Increased autonomy weakens the shared component and, in practice, can diminish—or even eliminate—the likelihood of cointegration across the set. Low correlation does not preclude cointegration, but it weakens the case for cointegration, i.e., a tight long-run equilibrium. This is tested for empirical evidence in the empirical section[4] Crises amplify comovement through two reinforcing channels. First, the variance channel operates when higher volatility in the common factors increases their share of total long-horizon variance. Second, the loading-alignment channel arises when lower dispersion in $\psi_{ik,r}$ makes currency responses more similar in both magnitude and sign, reducing idiosyncratic offsets. Even without introducing new stochastic trends, these channels generate tighter long-run alignment—a pattern consistent with the "risk-on/risk-off" behavior observed during global financial stress.

Proposition 3. *If new* ***persistent global shocks*** *emerge, or if* ***autonomy shifts*** *driven by, for example, the emergence of a never-before-seen idiosyncratic stochastic trend, such that*

$$\mathrm{rank}(\Lambda_{crisis}) > \mathrm{rank}(\Lambda_{non\text{-}crisis}),$$

then the number of cointegrating relationships falls. In the extreme case where $\mathrm{rank}(\mathbf{\Lambda}_{crisis}) = N$*, cointegration disappears entirely.*

Certain crises—such as the COVID-19 pandemic—introduce structural changes that are *not spanned* by existing common stochastic trend global factors, such as pandemic-specific risk premia, supply-chain dislocations, or regime-specific commodity shocks. If these shocks are persistent, load heterogeneously across currencies, and are linearly independent of existing global factors, then they increase the effective dimension of the global factor process g_t, either raising the rank of Λ_r by introducing a new common stochastic trend, or/and increasing the degree of autonomy (for example, by introducing an idiosyncratic stochastic trend) so that all currencies don't all load the same. This weakens cointegration, and in the limiting case of full rank of common stochastic trend ($\mathrm{rank}(\Lambda_r) = N$), a complete breakdown of cointegration across currencies.

3.7. Theoretical background of long-run currency linkages

To connect the empirical framework to established international finance theory, this section outlines the key economic channels that motivate regime-dependent long-run currency linkages. Understanding the long-run comovement of major currencies requires grounding the analysis in the structural forces that connect global financial markets. Exchange rates reflect the interaction of global capital flows, cross-border portfolio allocation, arbitrage conditions, safe-asset demand, and real-side fundamentals. Together, these channels determine

[4] For formal inference, cointegration tests remain decisive; the correlation-based argument here is interpretive.

whether currencies share persistent stochastic components or retain structural autonomy across regimes. A large class of open-economy and international asset-pricing models predicts that deep, internationally traded currencies load on persistent global forces. World interest rates, global liquidity conditions, and the international demand for high-quality safe assets generate common exposures across reserve currencies. When investors rebalance globally diversified portfolios, capital flows concentrate in major currencies, creating a dominant stochastic component in exchange rate dynamics. In dynamic asset-pricing terms, these currencies share exposure to a global factor that reflects time-varying risk-bearing capacity and cross-border capital allocation. This mechanism provides the theoretical basis for testing whether major exchange rates collapse onto a common long-run trend. Arbitrage conditions further reinforce potential long-run linkages. Covered and uncovered interest parity connect currency returns through cross-market pricing relationships. Although short-run deviations from uncovered interest parity are well documented, persistent arbitrage activity, balance-sheet adjustments, and global hedging flows can induce stable long-run relationships among exchange rates. Dealer balance-sheet constraints, limits to arbitrage, and convenience yields on safe assets create channels through which currencies respond systematically to common global shocks. These arbitrage foundations justify the use of cointegration methods to identify permanent equilibrium relationships.

At the same time, currencies differ in their structural roles within the international monetary system. Safe-haven currencies such as the Japanese yen and Swiss franc attract capital during periods of heightened global uncertainty. Theoretical models of precautionary demand, liquidity preference, and safe-asset scarcity predict that these currencies may exhibit greater autonomy from global trends, as their valuations respond directly to shifts in global risk sentiment. Such mechanisms imply the possibility of long-run segmentation, particularly during crisis episodes. Real-side factors provide an additional source of structural differentiation. Commodity-linked currencies, such as the Australian dollar, derive part of their long-run valuation from persistent terms-of-trade shocks driven by global commodity demand. These real shocks evolve slowly and may generate independent stochastic components in exchange rate dynamics. As a result, commodity currencies can display partial autonomy even when short-run financial shocks are common across markets. Finally, crisis models predict that systemic events alter the relative importance of global and idiosyncratic forces. During episodes of severe financial stress, flight-to-liquidity effects, global deleveraging, and funding constraints amplify common global factors, potentially reducing long-run autonomy across currencies. However, these same mechanisms may reinforce the distinct role of safe-haven currencies. Theoretical considerations, therefore, imply that long-run exchange rate linkages are likely to be regime dependent, motivating the empirical strategy of estimating cointegration relationships separately across crisis episodes. Taken together, these channels imply that currencies may share persistent global components, differ systematically in their structural loadings, and exhibit regime-dependent long-run comovement. A cointegration-based framework provides the appropriate empirical tool for identifying such permanent relationships and evaluating how crisis conditions reshape the long-run structure of major exchange rates.

4. Data and methodology

We use daily spot exchange rates for the five most globally traded non-USD currencies: EUR, GBP, JPY, CHF, and AUD. All exchange rates are harmonized using a consistent quotation convention in which the foreign currency is the base currency and the USD is the quote currency. In standard FX notation, the first currency in a pair is the base currency and the second is the quote currency. Accordingly, each series is expressed as $\text{currency}_i/\text{USD}$ for $i \in \{\text{EUR}, \text{GBP}, \text{JPY}, \text{CHF}, \text{AUD}\}$ so that, for example, EUR/USD denotes the number of USD received for one euro. This harmonized convention ensures that movements in all currencies are expressed in comparable dollar terms, simplifying interpretation and preserving consistency throughout the empirical analysis. We use daily data because the research question concerns the stability of long-run equilibrium relationships during crisis periods that are characterized by extremely fast-moving financial conditions. Global crises, especially the Global Financial Crisis and the COVID-19 pandemic, often produce high-frequency market dislocations. Lower-frequency data (weekly or monthly) would compress these episodes, obscure the timing of shifts, and potentially mix fundamentally different crisis phases into a single observation. Daily frequency preserves the shifts that cointegration tests are sensitive to, improves the detection of regime-dependent stochastic trends, and aligns with the fact that exchange rates are continuously traded in the world's deepest financial market. This approach is consistent with FX cointegration studies that rely on daily data to capture long-run relations under high-frequency market adjustment.[5] We restrict the analysis to the five major non-USD currencies—EUR, GBP, JPY, CHF, and AUD—because they form the core of global FX liquidity and are the currencies most plausibly driven by common global stochastic trends. According to the 2025 BIS (Bank for International Settlements), Triennial Survey (Bank for International Settlements, 2025), these currencies account for the dominant share of worldwide FX turnover and represent the key structural categories in international finance. In particular, EUR and GBP are deep global reserve currencies; JPY and CHF are global safe-haven currencies; and AUD is a highly traded commodity-linked currency. These currencies exhibit stable liquidity across major crises and are precisely those for which long-run comovement reflects global, rather than region-specific, forces. This aligns with our objective of studying global stochastic trends and long-run currency dynamics across systemic events. To examine regime-dependent shifts in long-run currency comovement, we divide the sample into three crisis periods widely recognized in both the academic and policy literatures: the dot-com crash (May 1997–June 2000), the Global Financial Crisis (August 2007–June 2009), and the COVID-19 pandemic (February 2020–January 2022).

4.1. Criteria for Crisis Period Selection

This paper focuses on global systemic crisis episodes that are relevant for the long-run structure of foreign exchange markets. Our objective is not to study exchange-rate stress in isolation, but to examine how large-scale, system-wide shocks reshape the persistent stochastic trends, cointegration relationships, and long-horizon comovement among major currencies. Accordingly, a crisis period is defined as an episode that satisfies three criteria. First, the shock must be global or USD-centered in nature, given the central role of the USD in international funding markets, trade invoicing, and global risk pricing. Such shocks propagate broadly across financial markets and countries, affecting exchange rates through funding constraints, balance-sheet channels, and shifts in global risk premia rather than through localized currency-specific mechanisms. Second, the episode must be associated with a sustained and systemic repricing of global risk and liquidity conditions, rather than a short-lived market dislocation. Third, the crisis must plausibly introduce or alter persistent global stochastic

[5](See, e.g., Baillie and Bollerslev, 1989; Flood and Rose, 1995; Inoue and Rossi, 2012).

trends, making it relevant for the analysis of long-run cointegration and currency autonomy rather than purely short-run volatility dynamics. The dot-com crash, the Global Financial Crisis, and the COVID-19 pandemic satisfy these criteria. Although these episodes are often discussed in the context of equity or credit markets, they were accompanied by profound disruptions in global funding conditions, capital flows, and safe-haven demand, all of which are central transmission channels in foreign exchange markets. In particular, the Global Financial Crisis represented a global USD funding shock, while the COVID-19 pandemic triggered an unprecedented and coordinated global liquidity response, with large and persistent effects on exchange rates and international financial integration. By contrast, several well-known currency-related or regional crises are not included because they do not meet these criteria for the research question at hand. Episodes such as the European Sovereign Debt Crisis or the Asian Financial Crisis were primarily regional in scope or currency-bloc specific. These episodes are often analyzed as regionally concentrated shocks. Because our empirical design targets USD-based major exchange rates and asks whether global/USD-centered systemic events coincide with changes in the dimension of common long-run trends, we focus on crisis episodes widely viewed as global in scope.[6]

Insert Table 1

Insert Figure 2

4.2. Empirical Hypotheses

Our framework yields several predictions concerning the long-run structure of major exchange rates across crisis regimes. Each hypothesis is formulated as a statistical null hypothesis and is empirically evaluated using the Johansen cointegration tests, vector error correction model estimation, and long-horizon return correlation analysis described in the methodology section.

Hypothesis 1 (Existence of Cointegration)**.** *In every crisis regime, the cointegration rank equals zero.*

This hypothesis is tested using Johansen trace and maximum-eigenvalue statistics applied to the individual crisis period.

Hypothesis 2 (Crisis-Dependent Cointegration)**.** *A similar cointegration structure exists across all crisis regimes.*

This hypothesis is evaluated by comparing the Johansen trace and maximum-eigenvalue rank determinations across the dot-com crash, the Global Financial Crisis, and the COVID-19 pandemic subsamples.

Hypothesis 3 (Correlation Implication of Cointegration)**.** *Crisis regimes in which the null of no cointegration cannot be rejected exhibit low average long-horizon return correlations.*

[6] Applying the same framework to more regionally concentrated crises—such as the European Sovereign Debt Crisis or the Asian Financial Crisis—would be a natural and valuable extension.

This hypothesis is tested by computing average pairwise long-horizon return correlations across currencies within each crisis subsample and comparing these averages across regimes classified by their estimated cointegration rank.

Hypothesis 4 (Regime-Dependent Adjustment Dynamics)**.** *Vector error-correction coefficients give rise to adjustment mechanisms that do not differ across currencies or across crisis regimes.*

This hypothesis is tested by examining the statistical significance and magnitude of the estimated error correction coefficients across currencies.

4.3. Methodology

Below, we present the relevant empirical procedure that we employ in line with the questions we hope to answer in the paper. These empirical techniques are implemented on the data presented in the previous section. The empirical framework in this paper answers a fundamental question in international finance: how global crises alter the long-run equilibrium structure, stochastic trends, and autonomy of major currencies. Prior research focuses almost exclusively on short-run exchange rate dynamics—risk premia, UIP regressions, volatility responses, and high-frequency comovements. These approaches do not identify the persistent components that determine the dimensionality and long-horizon behavior of currency systems. Our modeling framework is constructed precisely to fill this gap. The first pillar of the model is the existence of long-run global stochastic trends. Theory predicts that deep, internationally traded currencies share exposure to persistent world factors driven by global liquidity, safe-asset demand, and cross-border risk flows. Such forces manifest not in short-run returns but in the long-horizon comovement of exchange rates. Detecting these permanent components requires a cointegration-based structure; neither regressions in changes nor short-run VAR frameworks can identify equilibrium stochastic trends or measure how many such trends exist. The second pillar concerns crisis-driven shifts in long-run integration. Systemic events—such as the dot-com crash, the Global Financial Crisis, and the COVID-19 pandemic—alter global balance sheets, risk-bearing capacity, and safe-haven demand. Theory predicts that these shocks may compress idiosyncratic fundamentals and increase reliance on global factors, thereby changing the long-run linkage among currencies. Our empirical model explicitly estimates global trend loadings and autonomy measures separately across crises, allowing us to test whether long-run integration strengthens, weakens, or fragments during different systemic episodes. Standard TVP–VAR or quantile VAR methods capture evolving short-run dynamics but cannot detect structural changes in the permanent equilibrium relationships of the FX system. The third pillar derives from structural heterogeneity across currency types. Reserve currencies (EUR, GBP) are expected to load heavily on the global trend due to their role as funding and transaction units; safe-haven currencies (JPY, CHF) should retain greater autonomy due to independent liquidity demand; and commodity-linked currencies (AUD) should exhibit partial autonomy driven by persistent terms-of-trade cycles.

These theoretical distinctions manifest primarily in long-run trend behavior, not in short-run return responses. A cointegration-based model allows these roles to appear in the estimated long-run loadings and autonomy metrics, whereas short-run VARs impose symmetric or near-symmetric dynamics across currencies. Taken together, these considerations justify the empirical modeling approach adopted in the paper. Our framework provides a comparative advantage in three ways. First, it identifies and estimates the dominant global stochastic trends that anchor long-run currency behavior. Second, it separates permanent from transitory components, enabling a clean measurement of long-run autonomy. Third, it allows crisis-specific and currency-specific heterogeneity to emerge naturally in the long-run structure of the FX system. In contrast, TVP–VARs, Q–VARs, and related short-run models are unable to estimate persistent equilibrium components, long-run dimensionality, or the crisis-dependent behavior of global trends. For these reasons, the cointegration-based empirical model is not only methodologically appropriate but theoretically necessary for answering the long-horizon questions at the core of the paper. It directly targets the missing dimension in the literature: whether global crises alter the permanent structure of the FX system or merely generate temporary volatility. Below, we describe the empirical procedure.

We begin by verifying the order of integration of each exchange rate series using the Augmented Dickey-Fuller and Phillips-Perron tests. We include intercepts (and deterministic trends when appropriate) to test for unit roots and confirm that the series are nonstationary and integrated. We also confirm that the series are nonstationary and integrated of order one by testing that the first differences are stationary. This ensures that all series share the same integration order, a precondition for Johansen cointegration analysis. For both the Augmented Dickey-Fuller and Phillips-Perron tests, the null hypothesis is the presence of a unit root (nonstationarity), while the alternative is stationarity. Tests are implemented in levels and first differences, and the resulting statistics are reported. Having established the integration properties of the data, we apply the Johansen (1988, 1991) cointegration procedure to examine whether stable long-run equilibrium relationships exist among the $N = 5$ major currencies. The vector error correction representation is:

$$\Delta Y_t = \mu + \Pi Y_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta Y_{t-i} + \varepsilon_t, \tag{15}$$

where Y_t is the $N \times 1$ vector of log spot rates, Γ_i capture short-run dynamics, and $\Pi = \alpha\beta'$ captures long-run equilibrium relations. Here, β contains the cointegrating vectors, while α contains the speed-of-adjustment coefficients. If $r = \text{rank}(\Pi) = 0$, no long-run equilibrium exists; if $0 < r < N$, then r distinct cointegrating relationships characterize long-run dynamics. Johansen proposes two likelihood-ratio statistics to determine r. The trace test, which evaluates the null hypothesis of at most r cointegrating relations against the alternative of more than r:

$$\lambda_{\text{trace}}(r) = -T \sum_{i=r+1}^{N} \ln(1 - \hat{\lambda}_i), \tag{16}$$

where T is the sample size and $\hat{\lambda}_i$ are the estimated eigenvalues, ordered as $\hat{\lambda}_1 > \hat{\lambda}_2 > \cdots > \hat{\lambda}_N$. The maximum-eigenvalue test, which evaluates the null of exactly r cointegrating relations against the alternative of $r + 1$:

$$\lambda_{\max}(r, r+1) = -T \ln(1 - \hat{\lambda}_{r+1}). \tag{17}$$

Both tests are widely used in empirical finance applications. However, Lütkepohl et al. (2001) shows that the trace statistic generally exhibits superior power in small and moderate samples. Since our segmented crisis-period datasets are not especially large, we therefore base most of our cointegration conclusions on the trace statistic. Johansen's procedure is invariant to the choice of normalization: selecting a variable to normalize on only affects presentation of the cointegrating vector, not the rank. In our analysis, we normalize on the euro (EUR/USD). The euro is the most liquid and least policy-managed non-USD currency, making it a natural benchmark for global FX linkages. Alternative numeraires (GBP, AUD, CHF, JPY) are more prone to idiosyncratic or regime-specific shocks, which could bias the interpretation of the vectors. When cointegration is established, we estimate a vector error correction model to quantify the strength of the equilibrium relationships. The speed of adjustment coefficients in α measures how quickly deviations from long-run equilibrium are corrected. A larger (in absolute value) coefficient signals a stronger restoring force toward equilibrium, which we interpret as evidence of robust cointegration dynamics.

To capture broader patterns of dependence that may arise even in the absence of cointegration (i.e., without common stochastic trends binding currencies together), we compute rolling returns for each currency over horizons of one day, one week, one month, and one quarter. We then examine correlations across the five currency pairs to assess the strength of their linkages at different horizons. As a parsimonious measure of the overall degree of dependence among the five currencies, we compute the simple average of all pairwise correlations of their returns. Let $R_{i,t}$ denote the return of currency i at horizon h (where $i = 1, \ldots, 5$), and let ρ_{ij} denote the sample correlation coefficient between $R_{i,t}$ and $R_{j,t}$ for $i \neq j$. The total number of distinct pairwise correlations is $\binom{5}{2} = 10$. The average correlation is then defined as $\bar{\rho} = \frac{1}{\binom{5}{2}} \sum_{i<j} \rho_{ij}$ which corresponds to the mean off-diagonal element of the 5×5 correlation matrix of returns. A higher value of $\bar{\rho}$ indicates stronger average linkages among the currencies, consistent with greater market integration or contagion, while a lower value implies weaker connections and greater potential for diversification. Although this measure averages across heterogeneous bilateral relationships, it provides a tractable single-index summary of systemic currency return dependence. We track $\bar{\rho}$ across horizons and crisis periods to assess how exchange rate comovement evolves. While cointegration analysis and correlation measures may appear similar in spirit, they capture distinct aspects of exchange rate dependence. Cointegration tests whether currencies share common stochastic trends, reflecting long-run equilibrium linkages that are stable over time. In contrast, correlations measure the strength of short-run comovements in returns across different horizons, regardless of whether such movements are anchored by an equilibrium relation. It is therefore possible for currencies to be highly correlated on average without being cointegrated, or conversely, to exhibit cointegration without strong short-run correlation. By employing both approaches, we obtain a richer characterization of exchange rate interdependence that distinguishes structural long-run integration from episodic short-run dependence, particularly across different crisis regimes.

5. Empirical results

We now turn to the empirical analysis, which directly tests the implications of our model. Proposition 2 states that if currencies load similarly on global common factors (i.e., they are cointegrated) and crisis volatility is elevated, then long-horizon return correlations must converge toward unity. By logical contrapositive, if correlations do not approach one, then both conditions cannot hold simultaneously; at least one of the required conditions must be violated.

Since crisis volatility almost always rises, the implication simplifies to the idea that if correlations do not become high enough to have magnitudes around one, then the currencies do not load similarly on the common factors (i.e., cointegration is destroyed and autonomy elevated). The logical derivation of this implication is set out in the Appendix.

Insert Figure 3

This provides a clear empirical test. If correlations remain modest so that we do not observe long-horizon correlations that approach one, then we conclude that autonomy persists and cointegration weakens. We evaluate these predictions using daily spot exchange rates across the three crisis subperiods. The results are presented in three stages. First, we establish the time-series properties of the data with unit root tests to confirm suitability for cointegration analysis. Second, we apply Johansen's procedure to examine the existence and strength of long-run cointegration relationships among the USD currency pairs during each crisis episode, estimating vector error correction models (where we find cointegration) to characterize the adjustment dynamics. Finally, we complement this analysis with rolling long-horizon return correlations, which provide a non-cointegration-based measure of systemic dependence and allow us to assess whether observed correlations approach the levels predicted by the model under crisis regimes. Together, these results provide an empirical test of the model's implications regarding crisis-driven convergence versus persistent currency autonomy.

Insert Table 2

Insert Table 3

Insert Table 4

Insert Table 5

Insert Table 6

Insert Table 7

Insert Table 8

Insert Table 9

Insert Table 10

In cases where we find sufficient evidence for cointegration, we analyze the associated adjustment dynamics using an error-correction framework as we document below.

5.1. Error–correction representation (Global Financial Crisis episode, $r = 1$)

With one cointegrating vector during the Global Financial Crisis, we normalize on EUR/USD and write

$$\begin{aligned} u_t = \mathrm{EUR/USD}_t - \beta_0 - \beta_1\,\mathrm{AUD/USD}_t - \beta_2\,\mathrm{CHF/USD}_t \\ - \beta_3\,\mathrm{GBP/USD}_t - \beta_4\,\mathrm{JPY/USD}_t, \end{aligned} \tag{18}$$

where u_t is the equilibrium error. In the long–run equilibrium $u_t = 0$; away from equilibrium $u_t \neq 0$ captures over/undervaluation of the normalized rate relative to the linear combination of the others. Let $\mathbf{s}_t = (\mathrm{EUR}, \mathrm{AUD}, \mathrm{CHF}, \mathrm{GBP}, \mathrm{JPY})'_t$ denote the five USD exchange rates and let p be the lag length in the underlying vector autoregression. The vector error correction model is

$$\Delta \mathbf{s}_t = \boldsymbol{\alpha}\, u_{t-1} + \sum_{j=1}^{p-1} \boldsymbol{\Gamma}_j\, \Delta \mathbf{s}_{t-j} + \boldsymbol{\delta} + \boldsymbol{\varepsilon}_t, \qquad \boldsymbol{\varepsilon}_t \sim \text{i.i.d.}(0, \boldsymbol{\Sigma}), \tag{19}$$

with $\boldsymbol{\alpha} = (\alpha_{\text{EUR}}, \alpha_{\text{AUD}}, \alpha_{\text{CHF}}, \alpha_{\text{GBP}}, \alpha_{\text{JPY}})'$ the speeds of adjustment and u_{t-1} is the lagged error–correction term. Expanding equation–by–equation gives the five short–run regressions that capture the short-run adjustment mechanism to the long-run linkages, wherever such linkages exist:

$$\begin{aligned}\Delta\text{EUR}_t = \alpha_{\text{EUR}}\, u_{t-1} + \sum_{j=1}^{p-1} \Big(&\gamma^{\text{EUR}}_{\text{EUR},j}\, \Delta\text{EUR}_{t-j} + \gamma^{\text{EUR}}_{\text{AUD},j}\, \Delta\text{AUD}_{t-j} + \gamma^{\text{EUR}}_{\text{CHF},j}\, \Delta\text{CHF}_{t-j} \\ &+ \gamma^{\text{EUR}}_{\text{GBP},j}\, \Delta\text{GBP}_{t-j} + \gamma^{\text{EUR}}_{\text{JPY},j}\, \Delta\text{JPY}_{t-j}\Big) + \delta_{\text{EUR}} + \varepsilon_{\text{EUR},t}.\end{aligned} \tag{20}$$

$$\begin{aligned}\Delta\text{AUD}_t = \alpha_{\text{AUD}}\, u_{t-1} + \sum_{j=1}^{p-1} \Big(&\gamma^{\text{AUD}}_{\text{EUR},j}\, \Delta\text{EUR}_{t-j} + \gamma^{\text{AUD}}_{\text{AUD},j}\, \Delta\text{AUD}_{t-j} + \gamma^{\text{AUD}}_{\text{CHF},j}\, \Delta\text{CHF}_{t-j} \\ &+ \gamma^{\text{AUD}}_{\text{GBP},j}\, \Delta\text{GBP}_{t-j} + \gamma^{\text{AUD}}_{\text{JPY},j}\, \Delta\text{JPY}_{t-j}\Big) + \delta_{\text{AUD}} + \varepsilon_{\text{AUD},t}.\end{aligned} \tag{21}$$

$$\begin{aligned}\Delta\text{CHF}_t = \alpha_{\text{CHF}}\, u_{t-1} + \sum_{j=1}^{p-1} \Big(&\gamma^{\text{CHF}}_{\text{EUR},j}\, \Delta\text{EUR}_{t-j} + \gamma^{\text{CHF}}_{\text{AUD},j}\, \Delta\text{AUD}_{t-j} + \gamma^{\text{CHF}}_{\text{CHF},j}\, \Delta\text{CHF}_{t-j} \\ &+ \gamma^{\text{CHF}}_{\text{GBP},j}\, \Delta\text{GBP}_{t-j} + \gamma^{\text{CHF}}_{\text{JPY},j}\, \Delta\text{JPY}_{t-j}\Big) + \delta_{\text{CHF}} + \varepsilon_{\text{CHF},t}.\end{aligned} \tag{22}$$

$$\begin{aligned}\Delta\text{GBP}_t = \alpha_{\text{GBP}}\, u_{t-1} + \sum_{j=1}^{p-1} \Big(&\gamma^{\text{GBP}}_{\text{EUR},j}\, \Delta\text{EUR}_{t-j} + \gamma^{\text{GBP}}_{\text{AUD},j}\, \Delta\text{AUD}_{t-j} + \gamma^{\text{GBP}}_{\text{CHF},j}\, \Delta\text{CHF}_{t-j} \\ &+ \gamma^{\text{GBP}}_{\text{GBP},j}\, \Delta\text{GBP}_{t-j} + \gamma^{\text{GBP}}_{\text{JPY},j}\, \Delta\text{JPY}_{t-j}\Big) + \delta_{\text{GBP}} + \varepsilon_{\text{GBP},t}.\end{aligned} \tag{23}$$

$$\begin{aligned}\Delta\text{JPY}_t = \alpha_{\text{JPY}}\, u_{t-1} + \sum_{j=1}^{p-1} \Big(&\gamma^{\text{JPY}}_{\text{EUR},j}\, \Delta\text{EUR}_{t-j} + \gamma^{\text{JPY}}_{\text{AUD},j}\, \Delta\text{AUD}_{t-j} + \gamma^{\text{JPY}}_{\text{CHF},j}\, \Delta\text{CHF}_{t-j} \\ &+ \gamma^{\text{JPY}}_{\text{GBP},j}\, \Delta\text{GBP}_{t-j} + \gamma^{\text{JPY}}_{\text{JPY},j}\, \Delta\text{JPY}_{t-j}\Big) + \delta_{\text{JPY}} + \varepsilon_{\text{JPY},t}.\end{aligned} \tag{24}$$

The coefficients α_i are the *speeds of adjustment.* They must be statistically significant and of signs consistent with error correction to ensure stability. With the above normalization,

$$u_{t-1} > 0 \quad \Longrightarrow \quad \text{EUR is too high and/or (AUD, CHF, GBP, JPY) are too low.}$$

Restoration of equilibrium then requires either

$$\Delta\text{EUR}_t < 0 \quad (\alpha_{\text{EUR}} < 0)$$

and/or

$$\Delta\mathrm{AUD}_t,\ \Delta\mathrm{CHF}_t,\ \Delta\mathrm{GBP}_t,\ \Delta\mathrm{JPY}_t > 0 \quad (\alpha_{\mathrm{AUD}},\ \alpha_{\mathrm{CHF}},\ \alpha_{\mathrm{GBP}},\ \alpha_{\mathrm{JPY}} > 0).$$

If a given α_i is (statistically) zero, currency i is weakly exogenous with respect to the long–run relation: it does not bear the burden of adjustment, while the currencies with significant α's do. The matrices $\mathbf{\Gamma}_j$ capture short–run propagation via lagged differences, and $\boldsymbol{\delta}$ collects deterministic terms (e.g., an unrestricted constant) as appropriate to the chosen specification. We chose an appropriate lag of $p = 2$, following the standard information criterion and the lag selection test, which was the lag used to perform the Johansen test described earlier. Below in Table 11, Table 12, and Figure 4, we present the long-run equilibrium relationship and the short-run dynamics.

Insert Table 11

Insert Table 12

Insert Figure 4

Figure 4 shows the estimated error–correction speeds (α_i) from the vector error correction model during the GFC episode. EUR, CHF, and GBP have statistically significant coefficients of roughly 0.03, 0.05, and 0.02, indicating that these currencies bear the main burden of restoring the long-run equilibrium. AUD and JPY, by contrast, are insignificant. The yen's coefficient is numerically large (0.689) but highly imprecise, consistent with its weakly exogenous behavior and safe-haven status; the diagram therefore represents AUD and JPY with thin arrows. Overall, the visualization confirms the regression evidence: adjustment was concentrated in the European majors, while AUD and especially JPY remained largely outside the error–correction mechanism.

5.2. Cointegration and adjustment (Global Financial Crisis vs. dot-com crash/COVID-19 pandemic)

During the Global Financial Crisis we find one cointegrating relation among USD quotes of the majors. The long-run combination loads positively on AUD/CHF/GBP and (slightly) negatively on JPY (up to normalization), and deviations are corrected primarily through EUR, CHF, and GBP with daily half-lives of roughly 2–5 weeks. The dot-com crash and the COVID-19 pandemic show no cointegration. These results align with our theory: in systemic USD-centered stress (Global Financial Crisis), autonomy compresses, comovement strengthens, and a stable anchor emerges; when shocks are more heterogeneous (the dot-com crash, the COVID-19 pandemic), autonomy rises and the long-run link disappears.[7]

Insert Figure 5

Turning now to the adjustment speeds, the vector error correction model coefficients are: $\alpha_{\mathrm{EUR}} = 0.030^{***}$, $\alpha_{\mathrm{CHF}} = 0.050^{***}$, $\alpha_{\mathrm{GBP}} = 0.020^{***}$, $\alpha_{\mathrm{AUD}} = -0.016\,(\mathrm{ns})$, and $\alpha_{\mathrm{JPY}} =$

[7]With the vector normalized on EUR/USD, estimated coefficients are $\{\beta_{\mathrm{AUD}}, \beta_{\mathrm{CHF}}, \beta_{\mathrm{GBP}}, \beta_{\mathrm{JPY}}\} = \{-0.834, -1.686, -0.374, 0.009\}$. Multiplying the whole vector by -1 gives the equivalent "positive loadings on AUD/CHF/GBP, slight negative on JPY" phrasing; the economic content is invariant to this normalization.

0.689 (ns). The coefficients suggest that EUR, CHF, and GBP adjust significantly to deviations from the long-run relation; AUD and especially JPY do not. In the Global Financial Crisis, the long-run cointegrating relationship is therefore restored primarily by EUR, CHF, and GBP, with JPY behaving as weakly exogenous to the cointegrating space. The half-lives associated with these adjustment coefficients, using the estimate $HL \approx \ln(2)/|\alpha|$ at a daily frequency, are as follows: CHF $\approx$ 14 days ($\alpha \approx 0.05$), EUR $\approx$ 23 days ($\alpha \approx 0.03$), and GBP $\approx$ 35 days ($\alpha \approx 0.02$). These magnitudes are consistent with gradual but economically meaningful equilibrium correction in high-liquidity foreign exchange markets.

5.3. Link to the propositions

The Global Financial Crisis results provide an instance of autonomy compression: a strong common USD factor and similar loadings across several majors yield (i) evidence of cointegration and (ii) plausible error-correction in the non-yen pairs. By contrast, the dot-com crash and the COVID-19 pandemic show no cointegration, consistent with heterogeneous responses or higher autonomy for the currencies. Table 11 reports the vector error correction model results during the Global Financial Crisis. The error-correction term is strongly significant in the EUR, CHF, and GBP equations, but insignificant for AUD and JPY. Thus, in the short run, deviations from the unique long-run relation are corrected primarily through the European majors, while AUD and (especially) JPY behave as weakly exogenous to the cointegrating space. The magnitudes imply economically meaningful convergence at a daily frequency: estimated half-life is approximately two weeks for CHF, three weeks for EUR, and five weeks for GBP.[8] Cross-currency spillovers align with standard risk/safe-haven channels. A stronger CHF two days ago predicts weaker AUD today ($\Delta \mathrm{CHF}_{t-2} < 0$ in the AUD equation), consistent with safe-haven strength foreshadowing a retreat in a high-beta currency. Past risk-on moves in AUD precede subsequent JPY underperformance ($\Delta \mathrm{AUD}_{t-1} < 0$ in the JPY equation), consistent with the yen's safe-haven profile. Sterling transmits to continental Europe with a short delay: $\Delta \mathrm{GBP}_{t-2}$ significantly raises both ΔEUR and ΔCHF. Conversely, JPY pressure spills into GBP, with both $\Delta \mathrm{JPY}_{t-1}$ and $\Delta \mathrm{JPY}_{t-2}$ negative in the GBP equation.

This concentration of adjustment on European majors, with AUD and JPY playing distinct risk and safe-haven roles, dovetails with our regime-based interpretation. During globally systemic, USD-centered stress—such as the GFC—a stable long-run anchor emerges, and short-run re-equilibration is executed by the currencies most tightly linked to that anchor. There was no long-run equilibrium relationship to adjust to (i.e., no cointegration) during either the dot-com crash or the COVID-19 pandemic. While the dot-com crash was USD-centered, it was not globally systemic. Conversely, the COVID-19 pandemic was globally systemic, but not USD-centered. Our results suggest that for a crisis to coincide with cointegration among major dollar FX pairs, it must be both globally systemic and USD-centered; one condition alone is not sufficient. The Global Financial Crisis episode satisfies both conditions as it was not only globally systemic, but it also originated from the U.S., making it USD-centered. Our evidence is consistent with (though cannot prove from three episodes alone) the idea that cointegrating links among major USD pairs tend to appear when stress is both globally systemic and USD-centered—as in the Global Financial Crisis.

[8] Half-life $\approx \ln(0.5)/\ln(1-|\alpha|)$. The sign of α depends on the normalization of the cointegrating vector; economic "speed" is governed by $|\alpha|$.

5.4. Correlation of (long-horizon) rolling returns

Insert Table 13

Insert Table 14

Insert Table 15

The average correlations of H-horizon currency returns, where foreign exchange is expressed as the value of 1 unit currency in USD, are shown in Table 13, 14, and 15. The horizon correlations mostly fall in the weak-to-moderate range (0.1–0.4 in absolute value), often near zero and sometimes negative, and overall mostly remain far from one. This indicates that currencies do not move in a tightly synchronized way in response to common USD shocks. Interpreted through Proposition 2, this pattern is consistent with higher autonomy and, accordingly, a reduced scope for long-run links, matching the lack of cointegration in the dot-com crash and the COVID-19 pandemic. By contrast, during the GFC, we recovered a single cointegrating relation. That the cointegration rank, i.e., the number of cointegration relations, equals $1 < 4$ (the $N-1$ maximum for $N = 5$ USD quotes) shows that meaningful cross-currency heterogeneity persists, which is why a richer set of cointegrating relations does not materialize. A critical clarification is warranted on the distinction between visual inspection, correlation, and cointegration in our analysis. While graphical inspection of exchange rate series may offer intuition, visual impressions are not a reliable basis for inference. Most financial time series do not display relationships that can be reliably inferred by eye. Statistical measures are indispensable: to establish whether currencies co-move strongly, one must compute formal correlation measures. Without such metrics, inference risks being misleading. Our results confirm that correlations among long-horizon currency *returns* mostly lie below 0.4 in magnitude, with the bulk of values on the weak side. Crucially, these correlations must be computed on returns rather than levels, since nonstationary levels would spuriously inflate the correlation. By contrast, cointegration analysis is necessarily performed on *levels*, because long-run equilibrium concerns the presence of common stochastic trends in the underlying processes, not short-run fluctuations. Thus, correlation and cointegration are conceptually distinct: correlation measures short-run comovement in returns, while cointegration tests for shared stochastic trends in levels.[9]

This distinction underpins Proposition 2. If currencies load similarly on global stochastic trends, then long-horizon return correlations should converge toward one. The contrapositive is equally informative: if correlations do *not* converge, then loadings must be heterogeneous or additional independent stochastic trends remain. Our evidence aligns partly in the case of the Global Financial Crisis, and fully in the cases of the dot-com crash and the COVID-19 pandemic, with this logic, consistent with the contrapositive interpretation of Proposition 2. During the **dot-com crash** and **the COVID-19 pandemic**, Johansen tests indicate no cointegration (rank = 0), implying full autonomy among currencies with no common stochastic trends. In these regimes, long-horizon correlations fail to converge to one, exactly as Proposition 2 predicts. During the **Global Financial Crisis**, by contrast, we find one cointegrating relation (rank = 1 of a possible four), implying partial commonality: some stochastic

[9] As a simple example, two independent random walks with drift (say, EUR/USD and GBP/USD) will often show high correlations in levels purely because both trend upward over time. Their returns, however, are uncorrelated. This illustrates why correlations on levels are misleading and why cointegration is the correct tool for testing shared long-run trends.

trends were shared, others idiosyncratic. Correspondingly, long-horizon correlations exhibit only partial alignment, again consistent with the proposition. To visualize this implication, Figures 6–8 report rolling 6-month correlations of USD currency returns against EUR/USD during each crisis episode. These plots reveal whether correlations moved toward the theoretical full-convergence benchmark of +1 under elevated volatility (Note that EUR/USD itself is the base series, so its correlation with itself is identically +1; the dotted line thus serves only as a theoretical benchmark rather than an additional data series). In the **dot-com crash** and the **COVID-19 pandemic** panels, correlations remain volatile and often near zero or negative, with no line persistently approaching one. By contrast, during the **Global Financial Crisis**, correlations rise more strongly, and the EUR–GBP pair in particular stabilizes near 0.75–0.9. This distinctive elevation signals a partial but not universal alignment of currency dynamics.

Insert Figure 6

Insert Figure 7

Insert Figure 8

As a complementary robustness check, Figure 9 first reports the JP Morgan Global Currency Volatility Index (JPMVXYGL) with crisis windows highlighted, providing the underlying volatility benchmark. Building on this, Figure 10 then plots the same long-horizon rolling correlations against log(JPMVXYGL). This cross-sectional view directly tests Proposition 2: during the GFC, high volatility coincided with partial convergence, while in the dot-com crash and the COVID-19 pandemic, high volatility corresponded to dispersed or negative correlations, consistent with heterogeneous factor loadings and the absence of cointegration.

Insert Figure 9

Insert Figure 10

In sum, three points emerge: (i) correlations must be computed on returns, not levels; (ii) cointegration must be tested on levels, not returns; and (iii) correlation and cointegration are distinct and non-equivalent, with a lack of convergence in correlations providing evidence of heterogeneous loadings or additional independent trends.

5.5. Linking results to hypotheses

The Johansen trace and maximum-eigenvalue tests reject the null of no cointegration during the Global Financial Crisis, with an estimated rank of one, while the null cannot be rejected during the dot-com crash and the COVID-19 pandemic. Accordingly, Hypothesis 1, which posits zero cointegration rank in every crisis regime, is rejected. Hypothesis 2, which states that a similar cointegration structure exists across crisis regimes, is also rejected because the estimated rank differs across subsamples. These results demonstrate that the long-run equilibrium structure of major USD exchange rates is regime-dependent rather than invariant across crises. The existence of a cointegrating relation during the Global Financial Crisis, but not during the other two episodes, implies that systemic stress does not uniformly strengthen or weaken long-run linkages; instead, the dimensionality of common stochastic trends varies across crisis environments. The long-horizon returns correlation analysis provides some support for this regime dependence. Average long-horizon return correlations are low during the dot-com crash and the COVID-19 pandemic, where the hypothesis of no cointegration cannot be rejected. This pattern is consistent with Hypothesis 3, which predicts weaker long-horizon return comovement in regimes without detected cointegration. Finally, Hypothesis 4 is rejected because the estimated adjustment coefficients during the Global Financial Crisis differ across currencies, with EUR, CHF, and GBP exhibiting significant error-correction dynamics while AUD and JPY are weakly exogenous. Hypothesis 4 is also rejected along the regime dimension. Indeed, adjustment dynamics exist only during the GFC episode—where there is evidence of cointegration—and, by definition, cannot occur during the dot-com crash or the COVID-19 pandemic, where we find no evidence of cointegrating relations. This confirms that adjustment dynamics differ not only across currencies but across crisis regimes as well. Taken together, the evidence confirms that both the existence of long-run equilibrium relations and the associated adjustment dynamics vary systematically across crisis regimes.

5.6. Discussion and implications

This section interprets the empirical results in light of the existing literature and the theoretical framework developed earlier. The evidence reveals that long-run exchange rate linkages are strongly crisis-dependent and that the dimensionality of the global currency system changes across episodes. In particular, the Global Financial Crisis (GFC) strengthens long-run linkages, whereas the COVID-19 pandemic weakens them. The dot-com crash represents an intermediate case in which volatility rose, but persistent long-run integration did not materialize. The GFC exhibits a statistically significant cointegrating relationship across the major currencies examined. This finding is consistent with theories emphasizing USD funding stress, safe-asset demand, and intermediary balance-sheet constraints as mechanisms that compress cross-country autonomy and synchronize currency adjustments (Baba and Packer, 2009; McGuire and Von Peter, 2012; Du et al., 2018). During this episode, currencies appear to load more heavily and more similarly on a dominant global factor, reducing the number of independent stochastic trends and generating stable long-run cointegrating relations. This pattern aligns with prior empirical evidence documenting increased short-run spillovers during systemic crises (Melvin and Taylor, 2009; Aloui et al., 2014), but extends it by demonstrating that the strengthening of linkages persists at long horizons. In contrast, the COVID-19 pandemic displays a breakdown of cointegration and evidence consistent with an increase in the dimensionality of the system.

Although short-run volatility and spillovers intensified sharply in early 2020, the long-run analysis indicates that exchange rates did not converge toward a common stochastic trend. Instead, the data are consistent with the presence of additional persistent forces not spanned by pre-existing global factors. Potential mechanisms include public-health uncertainty, mobility restrictions, supply-chain fragmentation, and unprecedented policy interventions such as large-scale swap-line expansions. Unlike the GFC, which reinforced the dominance of the global dollar funding channel, the COVID-19 pandemic appears to have introduced heterogeneous macro-financial shocks that affected currencies asymmetrically. The result is weaker long-run integration despite elevated short-run volatility. The dot-com crash does not exhibit persistent cointegration, despite elevated volatility in global equity markets. This outcome is consistent with the interpretation of the dot-com crash as an equity-centered shock with limited direct impact on global funding conditions. Exchange rates during this period appear to have remained largely segmented, and no durable compression of autonomy parameters is observed. These empirical patterns map directly to the autonomy framework developed earlier.

When autonomy parameters are compressed, currencies load more uniformly on global risk factors, and the number of common stochastic trends declines. When autonomy increases or new persistent factors emerge, the system's dimensionality increases, and long-run cointegrating relationships weaken. The evidence suggests that crisis episodes differ not only in intensity but also in their structural impact on the global factor space. In the GFC, the dominant dollar funding channel reduced effective autonomy and strengthened cointegration. During the COVID-19 pandemic, heterogeneous shocks and policy responses expanded the factor space and weakened cointegration. Cross-sectional heterogeneity across currency types further supports this interpretation. Safe-haven currencies exhibit greater independence across regimes, consistent with persistent safety demand. Commodity-linked currencies display partial integration reflecting exposure to global demand cycles but retain idiosyncratic components linked to terms-of-trade dynamics. Reserve currencies show deeper integration during the GFC but regain independence when additional global forces emerge. These differences underscore that long-run linkages depend jointly on funding roles, factor exposures, and crisis-specific transmission mechanisms. The results carry broader implications. First, they challenge the common presumption that crises uniformly increase international financial integration. Instead, integration is regime-specific and depends on whether a crisis compresses the global factor structure or expands it. Second, they imply that the dimensionality of exchange rate systems is not constant over time. Models that assume a fixed number of global factors may therefore overlook important regime-dependent shifts in long-run dynamics. Finally, the findings suggest that policy interventions can alter the structure of global linkages, not merely short-run volatility patterns. Overall, the evidence indicates that foreign exchange markets adjust to crises through structural changes in their underlying factor composition. These adjustments differ meaningfully across episodes, implying that long-run exchange rate relationships cannot be assumed to behave uniformly across crises.

6. Conclusion

We show that long-run foreign exchange linkages during crises are regime-dependent: crises do not affect the long-run structure of major currencies uniformly. To rigorously demonstrate this point, we present a model that links cointegration outcomes to the rank of the pass-through matrix and predicts that cointegration is more likely to be present when currencies all respond similarly to a dominant global shock, whereas heterogeneous or novel persistent shocks can raise the effective trend dimension and dissolve long-run cointegrating relationships. Empirically, we apply Johansen rank tests and estimate a vector error correction model for five major exchange rates vis-à-vis the USD: EUR/USD, GBP/USD, JPY/USD, CHF/USD, and AUD/USD across three widely studied crises: the dot-com crash, the Global Financial Crisis, and the COVID-19 pandemic. We find evidence of a surviving long-run cointegration relationship only during the Global Financial Crisis, while cointegration is not supported during the dot-com crash or the COVID-19 pandemic. This pattern is consistent with the factor mapping of the model, in the sense that when stress is global and centered on the USD, currency pass-through becomes more convergent across countries, reducing autonomy and thus decreasing the effective number of trend dimensions, allowing for stable long-term anchoring. When shocks are more heterogeneous or when additional persistent global factors emerge, the number of stochastic trends increases, and cointegration relationships may break down. During the Global Financial Crisis regime, when cointegration relationships existed, the adjustment dynamics were mainly concentrated in the EUR, CHF, and GBP, which bore the brunt of error correction, with an average daily half-life of approximately 2–5 weeks. The AUD and especially the JPY exhibited weak exogeneity, consistent with their roles as risk and safe-haven assets, respectively.

A central implication is that the number of stochastic trends and the tightness of long-horizon correlation are distinct objects. Elevated comovement can arise even when long-run cointegrating relationships do not hold, because stronger common exposures can increase correlation without restoring cointegration. From a practical perspective, these results imply that diversification across major USD exchange rates can deteriorate sharply in USD-centered systemic stress, even if long-run anchors are absent in other crisis environments. Our framework yields testable implications: (i) stronger common-shock intensity and more aligned loadings should coincide with higher cointegration rank; (ii) heterogeneous or novel persistent shocks should lower cointegration rank. Our paper verifies these predictions using the world's most liquid currency set vis-à-vis the USD. Future research can build on our findings in several directions. First, allowing cointegration ranks and autonomy parameters to evolve continuously over time, rather than across pre-defined crises, would help identify whether structural shifts in long-run FX linkages arise gradually or abruptly. Second, extending the autonomy framework to emerging-market currencies would reveal whether the crisis-dependent patterns documented here generalize beyond the G5 majors. Third, linking long-run autonomy to indicators of global liquidity, cross-currency basis stress, and safe-asset demand could provide a more unified view of how global financial conditions shape the permanent structure of exchange rates. Developing a general equilibrium model of crisis-dependent global stochastic trends, e.g., the emergence of new persistent factors during events like the COVID-19 pandemic, offers a promising avenue for deepening the theoretical foundations of long-run currency linkages.

Finally, while this paper focuses on global and USD-centered systemic crises, an important avenue for future research is to extend the framework to more regionally concentrated or currency-specific crisis episodes. For example, applying the model to the European Sovereign Debt Crisis, the Asian Financial Crisis, or major emerging-market currency crises would provide valuable insights into how long-run exchange rate linkages evolve in more segmented financial environments. Such crises differ from the global events studied here in that their transmission mechanisms are more localized and may not generate new global stochastic trends; instead, they reshape regional factor structures or currency-bloc dynamics. Examining these episodes would allow future work to assess whether the crisis-dependent changes in cointegration and currency autonomy documented in this paper are specific to global systemic shocks or also arise in regional or currency-union contexts.

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A. Appendix

In the Appendix, we provide all the proofs of all technical results.

A.1. Proof of Proposition 1 (Cointegration rank $N-1$)

To prove Proposition 1, the following lemma is applied.

Lemma 2. *(Cointegration rank $N - \text{rank}(\Lambda_r)$) Given regime r and assumptions A1–A4, the cointegration rank of X_t is:*

$$\textit{Cointegration rank of } X_t = N - \text{rank}(\Lambda_r).$$

Proof. Consider the long-run factor structure:

$$X_t = \mu + \mathbf{\Lambda}_r g_t + \nu_t,$$

where $X_t \in \mathbb{R}^N$ denotes the vector of exchange rates (e.g., log bilateral exchange rates), $\mathbf{\Lambda}_r \in \mathbb{R}^{N \times K}$ is the regime-specific matrix of factor loadings, $g_t \in \mathbb{R}^K$ is a vector of global stochastic trends assumed to be integrated of order one, and ν_t is a stationary vector process. By assumption A2, the components of g_t are mutually uncorrelated and not cointegrated. Thus, g_t represents K independent stochastic trends. The nonstationarity in X_t arises only from the term $\mathbf{\Lambda}_r g_t$. Let $r := \text{rank}(\mathbf{\Lambda}_r) \leq \min(N, K)$. Then the image of g_t under $\mathbf{\Lambda}_r$ spans an r-dimensional subspace of $\mathbb{R}^N$. This implies that the nonstationary component of X_t lies entirely within an r-dimensional subspace. Let us now consider the existence of cointegrating vectors. A cointegrating vector $\beta \in \mathbb{R}^N$ is such that the linear combination $\beta^\top X_t$ is stationary:

$$\beta^\top X_t = \beta^\top \mu + \beta^\top \mathbf{\Lambda}_r g_t + \beta^\top \nu_t.$$

Since μ is constant and ν_t is stationary, the stationarity of $\beta^\top X_t$ depends entirely on whether $\beta^\top \mathbf{\Lambda}_r g_t$ is stationary. But g_t is $I(1)$, so $\beta^\top \mathbf{\Lambda}_r g_t$ is stationary if and only if:

$$\beta^\top \mathbf{\Lambda}_r = 0.$$

That is, cointegrating vectors lie in the left null space of $\mathbf{\Lambda}_r$. The dimension of this null space is:

$$\dim\left(\{\beta \in \mathbb{R}^N : \beta^\top \mathbf{\Lambda}_r = 0\}\right) = N - \text{rank}(\mathbf{\Lambda}_r).$$

Hence, the number of linearly independent cointegrating vectors, i.e., the cointegration rank of X_t, is:

$$\text{Cointegration rank} = N - \text{rank}(\mathbf{\Lambda}_r).$$

□

Proof of Proposition 1. For any $b \in \mathbb{R}^N$, note:

$$b'\boldsymbol{X}_t = b'\mu_r + (b'\mathbf{\Lambda}_r)\, g_t + b'\Gamma \boldsymbol{z}_t + b'\boldsymbol{u}_t.$$

Since $\boldsymbol{z}_t$ and $\boldsymbol{u}_t$ are stationary, $b'\Gamma\boldsymbol{z}_t + b'\boldsymbol{u}_t$ is stationary integrated of order zero. The only potentially nonstationary component is $(b'\mathbf{\Lambda}_r)g_t$, because g_t is integrated of order one and its components are assumed not to be cointegrated (so they represent independent stochastic trends). Thus, if $b'\mathbf{\Lambda}_r = 0$, then $b'\boldsymbol{X}_t$ consists of a constant plus stationary $I(0)$ components and is therefore $I(0)$. In contrast, if $b'\mathbf{\Lambda}_r \neq 0$, then $b'\boldsymbol{X}_t$ inherits an $I(1)$ component and is therefore $I(1)$. Hence, the cointegrating space is $\{b \in \mathbb{R}^N : b'\mathbf{\Lambda}_r = 0\}$, of dimension $N - 1$ whenever $\text{rank}(\mathbf{\Lambda}_r) = 1$. This follows from Lemma 2.

A.2. Proof of Proposition 2 (Crisis Comovement Amplification of Long-Horizon Currency Returns)

The proof of Proposition 2 is divided into three steps. We start by computing the variance of the long-run returns, then the factor share of the total variance, and conclude by computing the correlation between long-run returns. We begin with the factor model for currency i in regime r:

$$X_{i,t} = \mu_{i,r} + \boldsymbol{\lambda}_{i,r}^{\top}\boldsymbol{g}_t + \boldsymbol{\Gamma}_i^{\top}\boldsymbol{z}_t + u_{i,t},$$

where $\boldsymbol{g}_t$ are global stochastic trends (e.g., integrated), $\boldsymbol{z}_t$ are stationary domestic/regional factors, and $u_{i,t}$ is idiosyncratic noise.

Taking first differences:

$$\Delta X_{i,t} = \boldsymbol{\lambda}_{i,r}^{\top}\Delta\boldsymbol{g}_t + \Delta(\boldsymbol{\Gamma}_i^{\top}\boldsymbol{z}_t + u_{i,t}).$$

Define the cumulative return over horizon H as:

$$Y_{i,H} \equiv \sum_{h=1}^{H}\Delta X_{i,t+h} = \boldsymbol{\lambda}_{i,r}^{\top}(\boldsymbol{g}_{t+H} - \boldsymbol{g}_t) + \varepsilon_{i,H},$$

where

$$\varepsilon_{i,H} = \Gamma_i^{\top}(z_{t+H} - z_t) + (u_{i,t+H} - u_{i,t})$$

is a difference between stationary processes.

(Variance of the long-run return): Because g_t is a vector of random walks,

$$\mathrm{Var}(\boldsymbol{g}_{t+H} - \boldsymbol{g}_t) = H\boldsymbol{\Sigma}_{g,r}.$$

With $\varepsilon_{i,H}$ uncorrelated with $\boldsymbol{g}_t$, we obtain

$$\mathrm{Var}(Y_{i,H}) = H\boldsymbol{\lambda}_{i,r}^{\top}\boldsymbol{\Sigma}_{g,r}\boldsymbol{\lambda}_{i,r} + \mathrm{Var}(\varepsilon_{i,H}).$$

Since $\varepsilon_{i,H}$ is a *difference of stationary levels*, its variance is bounded:

$$\mathrm{Var}(\varepsilon_{i,H}) = O(1).$$

Hence

$$\mathrm{Var}(Y_{i,H}) = H\boldsymbol{\lambda}_{i,r}^{\top}\boldsymbol{\Sigma}_{g,r}\boldsymbol{\lambda}_{i,r} + O(1).$$

(Factor share of total variance): The share of total variance explained by global factors at horizon H is

$$\frac{\text{Factor variance}}{\text{Total variance}} = \frac{H\boldsymbol{\lambda}_{i,r}^{\top}\boldsymbol{\Sigma}_{g,r}\boldsymbol{\lambda}_{i,r}}{H\boldsymbol{\lambda}_{i,r}^{\top}\boldsymbol{\Sigma}_{g,r}\boldsymbol{\lambda}_{i,r} + O(1)} = \frac{\boldsymbol{\lambda}_{i,r}^{\top}\boldsymbol{\Sigma}_{g,r}\boldsymbol{\lambda}_{i,r}}{\boldsymbol{\lambda}_{i,r}^{\top}\boldsymbol{\Sigma}_{g,r}\boldsymbol{\lambda}_{i,r} + O(1/H)}.$$

Thus, as $\Sigma_{g,r}$ rises in crisis, this ratio rises; and as $H \to \infty$, the $O(1/H)$ term vanishes so the factor share tends to 1. Thus, the common global factors explain a larger fraction of the long-run variance. This is the *variance channel* of comovement amplification.

(Correlation between long-run returns): For any pair i, j, define:

$$\rho_{ij}^{(H)} = \frac{\operatorname{Cov}(Y_{i,H}, Y_{j,H})}{\sqrt{\operatorname{Var}(Y_{i,H}) \operatorname{Var}(Y_{j,H})}}.$$

We have

$$\operatorname{Cov}(Y_{i,H}, Y_{j,H}) = H\boldsymbol{\lambda}_{i,r}^{\top}\boldsymbol{\Sigma}_{g,r}\boldsymbol{\lambda}_{j,r} + O(1),$$
$$\operatorname{Var}(Y_{i,H}) = H\boldsymbol{\lambda}_{i,r}^{\top}\boldsymbol{\Sigma}_{g,r}\boldsymbol{\lambda}_{i,r} + O(1), \qquad \operatorname{Var}(Y_{j,H}) = H\boldsymbol{\lambda}_{j,r}^{\top}\boldsymbol{\Sigma}_{g,r}\boldsymbol{\lambda}_{j,r} + O(1).$$

Dividing numerator and denominator by H,

$$\rho_{ij}^{(H)} = \frac{\boldsymbol{\lambda}_{i,r}^{\top}\boldsymbol{\Sigma}_{g,r}\boldsymbol{\lambda}_{j,r} + O(1/H)}{\sqrt{(\boldsymbol{\lambda}_{i,r}^{\top}\boldsymbol{\Sigma}_{g,r}\boldsymbol{\lambda}_{i,r} + O(1/H))(\boldsymbol{\lambda}_{j,r}^{\top}\boldsymbol{\Sigma}_{g,r}\boldsymbol{\lambda}_{j,r} + O(1/H))}}.$$

As $H \to \infty$, the bounded stationary differences vanish:

$$\lim_{H\to\infty} \rho_{ij}^{(H)} = \frac{\boldsymbol{\lambda}_{i,r}^{\top}\boldsymbol{\Sigma}_{g,r}\boldsymbol{\lambda}_{j,r}}{\sqrt{(\boldsymbol{\lambda}_{i,r}^{\top}\boldsymbol{\Sigma}_{g,r}\boldsymbol{\lambda}_{i,r})(\boldsymbol{\lambda}_{j,r}^{\top}\boldsymbol{\Sigma}_{g,r}\boldsymbol{\lambda}_{j,r})}}.$$

This is the cosine of the angle between the vectors $\Sigma_{g,r}^{1/2}\boldsymbol{\lambda}_{i,r}$ and $\Sigma_{g,r}^{1/2}\boldsymbol{\lambda}_{j,r}$. If the factor loadings are aligned (i.e., $\boldsymbol{\lambda}_{i,r} \approx \boldsymbol{\lambda}_{j,r}$), then the limit approaches 1. Thus, greater similarity in how currencies respond to global shocks leads to stronger long-run correlation. This is the *loading-alignment channel*.

A.3. Proof of Proposition 3 (Rank Change in Crisis and Cointegration)

Consider the factor structure of currency returns before and during crisis regimes:

$$X_t = \mu_r + \Lambda_r g_t + \Gamma_r z_t + u_t,$$

where X_t is the $N \times 1$ vector of currency series, Λ_r is the $N \times M_r$ loading matrix on the persistent global factors g_t, g_t is an $M_r \times 1$ vector of persistent global factors (stochastic trends), $\Gamma_r z_t + u_t$ represents the stationary component of the system, and the regime $r \in \{$dot-com crash, Global Financial Crisis, COVID-19 pandemic$\}$ indexes the different crisis episodes.

(Relation between rank of Λ_r and cointegration): By standard results in cointegration theory (see (Johansen, 1991); (Stock and Watson, 1988)), the number of common stochastic trends driving X_t equals the rank of Λ_r, denoted $\operatorname{rank}(\Lambda_r) = K_r$, where $K_r := \operatorname{rank}(\boldsymbol{\Lambda}_r) \leq \min(N, M_r)$. The dimension of the cointegrating space, i.e., the number of stationary linear combinations of X_t, equals

$$\dim(\text{cointegrating space}) = N - K_r.$$

(Effect of new persistent global factors): Suppose that during a crisis, new persistent global factors emerge that are independent of the pre-crisis factors. Formally, the crisis regime factor vector can be written as

$$g_t^{\text{crisis}} = \begin{bmatrix} g_t^{\text{pre}} \\ g_t^{\text{new}} \end{bmatrix},$$

where $\dim(g_t^{\text{new}}) = K_{\text{new}} > 0$, and g_t^{new} is independent of g_t^{pre}. The corresponding loading matrix expands as

$$\Lambda_{\text{crisis}} = [\Lambda_{\text{pre}} \quad \Lambda_{\text{new}}],$$

with Λ_{new} loading persistently on the new factors. Because g_t^{new} is independent, the rank of Λ_{crisis} satisfies

$$\text{rank}(\Lambda_{\text{crisis}}) \geq \text{rank}(\Lambda_{\text{pre}}).$$

Thus,

$$K_{\text{crisis}} = \text{rank}(\Lambda_{\text{crisis}}) > \text{rank}(\Lambda_{\text{pre}}) = K_{\text{pre}}.$$

(Effect on cointegration relations): Since the number of cointegrating relations equals $N - K_r$, an increase in the rank K_r reduces the cointegration dimension:

$$N - K_{\text{crisis}} < N - K_{\text{pre}}.$$

In the extreme case where $M_{\text{crisis}} \geq N$ and Λ_{crisis} attains full row rank N,

$$K_{\text{crisis}} = N,$$

The number of cointegrating relations becomes zero, meaning no stationary linear combination exists - the system has no cointegration.

(Effect of autonomy shifts on rank): Alternatively, even without new factors, if autonomy shifts such that the loading matrix Λ changes rank (e.g., if some currency responses become linearly independent from others), then

$$\text{rank}(\Lambda_{\text{crisis}}) > \text{rank}(\Lambda_{\text{pre}}),$$

and the same conclusion on cointegration applies. Hence, the emergence of new persistent global factors or autonomy shifts that increase the rank of Λ during crisis reduces the number of cointegrating relations, and if the rank reaches N, cointegration disappears entirely.

A.4. Proof of Lemma 1

Rewrite the no-arbitrage condition as a log-normal moment. From (5) and $X_{i,t} - X_{i,t+1} = -\Delta X_{i,t+1}$,

$$1 = \mathbb{E}_t\Big[\, M_{t+1}\, e^{-\Delta X_{i,t+1}} \,\Big] = \mathbb{E}_t\big[\, \exp\big(\log M_{t+1} - \Delta X_{i,t+1} \big) \,\big].$$

Define

$$Y_{i,t+1} \;:=\; \log M_{t+1} - \Delta X_{i,t+1}.$$

By the conditions of the lemma, we know that $\log M_{t+1}$ is Gaussian and $\Delta X_{i,t+1}$ is an affine function of Gaussian shocks; hence $Y_{i,t+1}$ is Gaussian conditional on $\mathcal{F}_t$:

$$Y_{i,t+1} \mid \mathcal{F}_t \;\sim\; \mathcal{N}(\mu_{Y,i,t},\, v_{Y,i,t}).$$

Therefore

$$\mathbb{E}_t\Big[e^{Y_{i,t+1}}\Big] \;=\; \exp\Big(\mu_{Y,i,t} + \tfrac{1}{2} v_{Y,i,t} \Big) \;=\; 1. \tag{A.1}$$

We next compute $\mu_{Y,i,t}$ and $\upsilon_{Y,i,t}$ and note $a_{i,r}$. To do this, we again refer to the first assumption of the lemma and plug the expression for $\log M_{t+1}$ into $Y_{i,t+1}$:

$$Y_{i,t+1} = \underbrace{\big(-r_t - \tfrac{1}{2}\varsigma_r^2\big)}_{\text{deterministic at } t} \; - \xi_{t+1} \; - \; \Delta X_{i,t+1}.$$

Using

$$\Delta X_{i,t+1} = a_{i,r} + \lambda'_{i,r}\,\Delta g_{t+1} + \Gamma'_i\,\Delta z_{t+1} + \epsilon_{i,t+1},$$

we have

$$Y_{i,t+1} = -r_t - \tfrac{1}{2}\varsigma_r^2 \; - \; \xi_{t+1} \; - \; a_{i,r} - \lambda'_{i,r}\Delta g_{t+1} - \Gamma'_i\Delta z_{t+1} - \epsilon_{i,t+1}.$$

Using $\xi_{t+1} = \lambda'_r\Delta g_{t+1} + \upsilon_{t+1}$, we have

$$Y_{i,t+1} = -r_t - \tfrac{1}{2}\varsigma_r^2 - a_{i,r} \; - \; (\lambda'_r + \lambda'_{i,r})\Delta g_{t+1} \; - \; \Gamma'_i\Delta z_{t+1} \; - \; \upsilon_{t+1} \; - \; \epsilon_{i,t+1}.$$

Hence, the conditional mean and variance are

$$\begin{aligned}
\mu_{Y,i,t} &= -r_t - \tfrac{1}{2}\varsigma_r^2 - a_{i,r}, \\
\upsilon_{Y,i,t} &= \mathrm{Var}_t\big((\lambda'_r + \lambda'_{i,r})\Delta g_{t+1} + \Gamma'_i\Delta z_{t+1} + \upsilon_{t+1} + \epsilon_{i,t+1}\big).
\end{aligned}$$

Using the orthogonality of blocks in the lemma, this variance decomposes additively:

$$\upsilon_{Y,i,t} = (\lambda_r + \lambda_{i,r})'\Sigma_{\Delta g,r}\,(\lambda_r + \lambda_{i,r}) \; + \; \Gamma'_i\Sigma_{\Delta z}\,\Gamma_i \; + \; \sigma_\upsilon^2 \; + \; \sigma_{\epsilon_i}^2, \tag{A.2}$$

where $\Sigma_{\Delta g,r} = \mathrm{Var}_t(\Delta g_{t+1})$ (regime r), $\Sigma_{\Delta z} = \mathrm{Var}_t(\Delta z_{t+1})$, and $\sigma_\upsilon^2 = \mathrm{Var}_t(\upsilon_{t+1})$, $\sigma_{\epsilon_i}^2 = \mathrm{Var}_t(\epsilon_{i,t+1})$.

Imposing (A.1) gives

$$\mu_{Y,i,t} + \tfrac{1}{2}\upsilon_{Y,i,t} \; = \; 0 \quad \Longrightarrow \quad -r_t - \tfrac{1}{2}\varsigma_r^2 - a_{i,r} + \tfrac{1}{2}\upsilon_{Y,i,t} \; = \; 0.$$

Therefore, the drift in the difference equation is pinned down by

$$a_{i,r} = -r_t - \tfrac{1}{2}\varsigma_r^2 + \tfrac{1}{2}\upsilon_{Y,i,t}. \tag{A.3}$$

Equivalently,

$$a_{i,r} \; = \; r_t + \tfrac{1}{2}\varsigma_r^2 \; - \; \tfrac{1}{2}\,\mathrm{Var}_t\big((\lambda_r + \lambda_{i,r})'\Delta g_{t+1} + \Gamma'_i\Delta z_{t+1} + \upsilon_{t+1} + \epsilon_{i,t+1}\big),$$

where we used $\upsilon_{Y,i,t}$ from (A.2). This is the risk-adjusted drift restriction implied by no-arbitrage under log-normality. Recognizing the difference of the end-to-end levels is obtained by summing the difference equation over time to recover levels, we have, by summing (12) over $s = 0, \ldots, t-1$:

$$X_{i,t} - X_{i,0} = \sum_{s=0}^{t-1}\Delta X_{i,s+1} = \sum_{s=0}^{t-1} a_{i,r} + \lambda'_{i,r}\sum_{s=0}^{t-1}\Delta g_{s+1} + \Gamma'_i\sum_{s=0}^{t-1}\Delta z_{s+1} + \sum_{s=0}^{t-1}\epsilon_{i,s+1}.$$

Using telescoping sums and the fact that $\epsilon_{i,s+1} = \Delta u_{i,s+1}$,

$$X_{i,t} = X_{i,0} + a_{i,r}\, t + \lambda'_{i,r}\, g_t + \Gamma'_i\, z_t + u_{i,t} - u_{i,0}.$$

Collect the constants and deterministic terms (including $X_{i,0} - u_{i,0}$ and the deterministic trend component implied by $a_{i,r}$) into

$$\mu_{i,r} \;:=\; X_{i,0} - u_{i,0} \;+\; \underbrace{\big(\text{risk-adjusted drift term consistent with (A.3)}\big)}_{\text{absorbed into } \mu_{i,r}}.$$

Thus, we can rewrite the level as

$$X_{i,t} \;=\; \mu_{i,r} \;+\; \lambda'_{i,r}\, g_t \;+\; \Gamma'_i\, z_t \;+\; u_{i,t}. \tag{A.4}$$

By construction, $u_{i,t}$ is stationary ($I(0)$) and we know that, g_t is $I(1)$ and z_t is $I(0)$. Hence (A.4) is an affine (linear-in-state) representation with $I(1)$ driver g_t, stationary control z_t, and stationary remainder $u_{i,t}$. The deterministic term $\mu_{i,r}$ aggregates initial conditions and the risk-adjusted drift identified by (A.3). This proves the claim in Lemma 1.

Below, we show a detailed argument that $\mathbb{E}_t[M_{t+1}] = e^{-r_t}$.

To compute $\mathbb{E}_\mathbf{t}[\mathbf{M_{t+1}}] = \mathbf{e^{-r_t}}$. Assume the stochastic discount factor M_t follows

$$\frac{dM_t}{M_t} \;=\; -\, r_t\, dt \;-\; \Lambda_t^\top\, dW_t, \tag{A.5}$$

where r_t is the short rate, W_t is a k-dimensional Brownian motion, and $\Lambda_t \in \mathbb{R}^k$ collects the (vector) prices of risk. By Itô's formula,

$$d \log M_t \;=\; \frac{dM_t}{M_t} \;-\; \frac{1}{2}\left(\frac{dM_t}{M_t}\right)^2.$$

Using (A.5), note that $\left(\frac{dM_t}{M_t}\right)^2 = (\Lambda_t^\top dW_t)^2 = \|\Lambda_t\|^2\, dt$. Hence

$$d \log M_t \;=\; -\, r_t\, dt \;-\; \Lambda_t^\top dW_t \;-\; \tfrac{1}{2}\, \|\Lambda_t\|^2\, dt. \tag{A.6}$$

Integrate (A.6) from t to $t + \Delta$:

$$\log M_{t+\Delta} - \log M_t \;=\; -\int_t^{t+\Delta} r_s\, ds \;-\; \tfrac{1}{2}\int_t^{t+\Delta} \|\Lambda_s\|^2 ds \;-\; \int_t^{t+\Delta} \Lambda_s^\top\, dW_s. \tag{A.7}$$

Define the (martingale) shock and its conditional variance:

$$\xi_{t+\Delta} \;\equiv\; \int_t^{t+\Delta} \Lambda_s^\top\, dW_s, \qquad \varsigma^2 \;\equiv\; \int_t^{t+\Delta} \|\Lambda_s\|^2 ds.$$

Conditionally on $\mathcal{F}_t$, $\xi_{t+\Delta} \sim \mathcal{N}(0, \varsigma^2)$. Then (A.7) becomes

$$\log M_{t+\Delta} \;=\; \log M_t \;-\; \int_t^{t+\Delta} r_s \, ds \;-\; \tfrac{1}{2}\,\varsigma^2 \;-\; \xi_{t+\Delta}. \tag{A.8}$$

If we take $\Delta = 1$ and assume $r_s \equiv r_t$ and $\Lambda_s \equiv \Lambda_t$ on $[t, t+1]$, then

$$\int_t^{t+1} r_s \, ds = r_t, \qquad \varsigma^2 = \int_t^{t+1} \|\Lambda_s\|^2 ds = \|\Lambda_t\|^2 \;\equiv\; \varsigma_r^2,$$

and $\xi_{t+1} = \Lambda_t^\top (W_{t+1} - W_t)$. Normalizing $M_t = 1$ (so $\log M_t = 0$), (A.8) reduces to

$$\log M_{t+1} \;=\; -\,r_t \;-\; \tfrac{1}{2}\,\varsigma_r^2 \;-\; \xi_{t+1}, \tag{A.9}$$

with $\xi_{t+1} \,|\, \mathcal{F}_t \sim \mathcal{N}(0, \varsigma_r^2)$. Since $\xi_{t+1} \sim \mathcal{N}(0, \varsigma_r^2)$, $\mathbb{E}_t[e^{-\xi_{t+1}}] = \exp\left(\tfrac{1}{2}\varsigma_r^2\right)$. Therefore

$$\mathbb{E}_t[M_{t+1}] = \exp\big(-r_t - \tfrac{1}{2}\varsigma_r^2\big)\, \mathbb{E}_t[e^{-\xi_{t+1}}] = \exp(-r_t),$$

so $-\frac{1}{2}\varsigma_r^2$ is exactly the Jensen correction that pins $\mathbb{E}_t[M_{t+1}] = e^{-r_t}$.

H.5. Logical Structure of Proposition 2

For clarity, we set out the formal logic underlying Proposition 2 and its testable implication.

(1) *General Statement:* If (A and B) then C:

$$(A \wedge B) \;\;\Rightarrow\;\; C,$$

where A denotes that currencies load similarly on global common factors, B captures the rise in crisis volatility, and C reflects the convergence of long-horizon return correlations toward one.

(2) *Contrapositive:* By standard logic,

$$\neg C \;\;\Rightarrow\;\; \neg(A \wedge B).$$

$$\neg(A \wedge B) \;\;\Leftrightarrow\;\; (\neg A \vee \neg B).$$

(3) *Crisis Simplification:* In systemic crises, volatility almost always rises. Hence $\neg B$ is implausible. This reduces the contrapositive to

$$\neg C \;\;\Rightarrow\;\; \neg A.$$

(4) *Testable Implication:* If correlations approach one (C), this is consistent with A (currencies load similarly on common factors $\Rightarrow$ reduced autonomy). If correlations remain modest ($\neg C$), then $\neg A$ (factor loadings are not aligned $\Rightarrow$ autonomy persists, cointegration weakens). This provides the empirical testing rule applied in the main text.

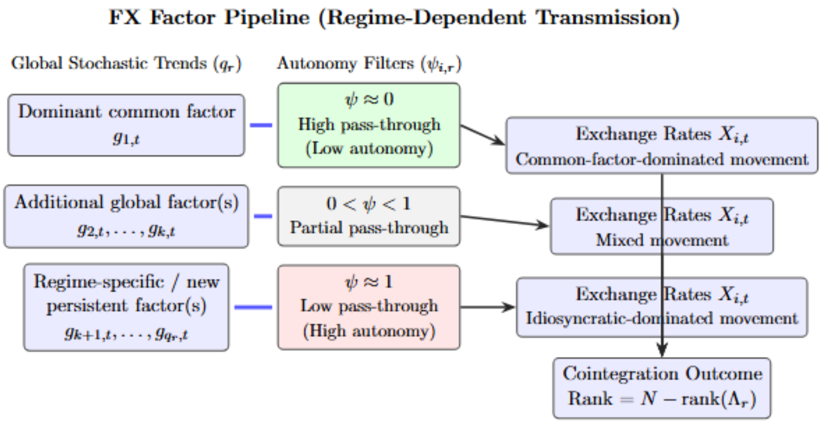

Figure 1: **Factor pipeline representation of exchange rate determination.** A regime-dependent set of global stochastic trends, $\{g_{1,t}, \ldots, g_{q_r,t}\}$, is transmitted through currency-specific autonomy filters, $\psi_{i,r}$, which regulate the extent of pass-through into exchange rates. Low autonomy leads to common-factor-dominated movements and stronger cointegration, while high autonomy results in idiosyncratic movements and weaker cointegration. Intermediate cases generate mixed dynamics.

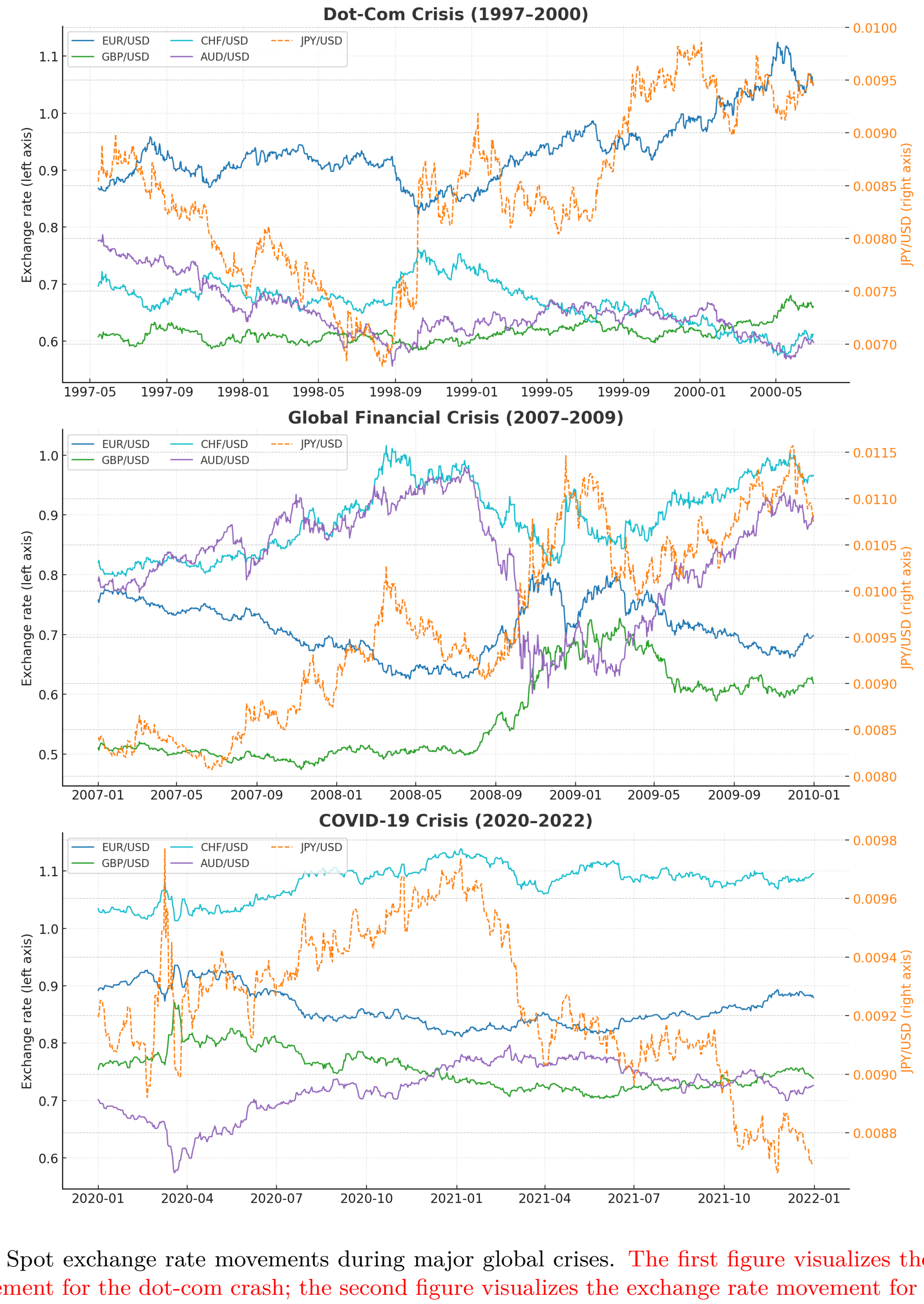

Figure 2: Spot exchange rate movements during major global crises. The first figure visualizes the exchange rate movement for the dot-com crash; the second figure visualizes the exchange rate movement for the Global Financial Crisis; and the third figure visualizes the exchange rate movement for the COVID-19 pandemic.

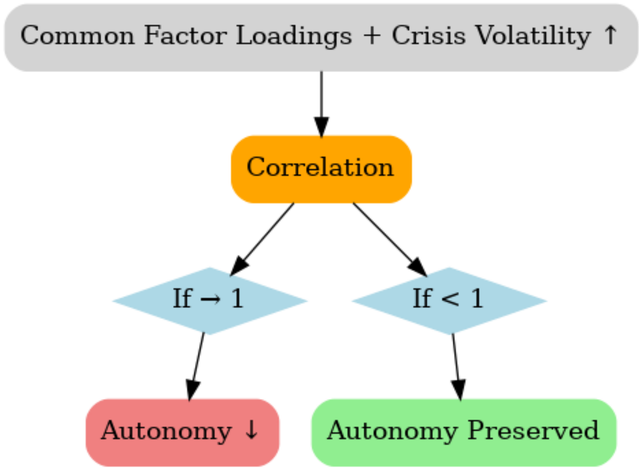

Figure 3: Stylized relationship between crisis volatility, currency loadings, and long-run correlation structure.

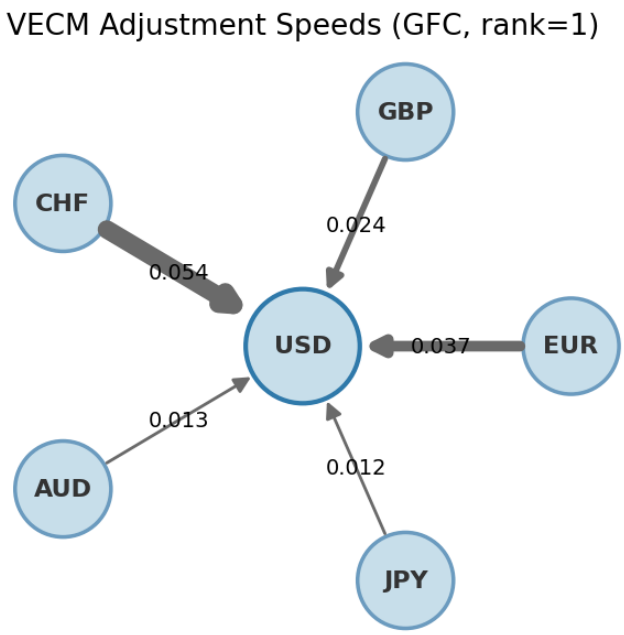

Figure 4: Estimated adjustment coefficients from the vector error correction model for major USD exchange rates during the Global Financial Crisis, assuming a cointegration rank of $r = 1$. The figure visualizes the estimated α_i coefficients associated with the error-correction term for each currency, with arrow thickness proportional to the absolute magnitude of the corresponding coefficient. The estimates are based on daily spot exchange rate data spanning 2007–2009.

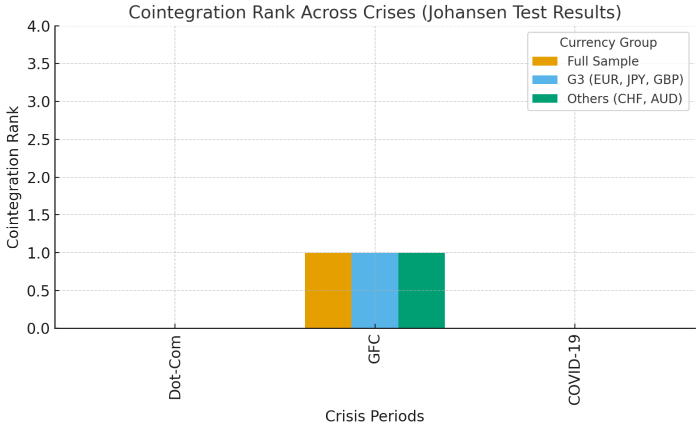

Figure 5: Estimated cointegration rank for major exchange rates across crisis periods based on Johansen trace test results. The figure reports the inferred cointegration rank for the dot-com crash (1997–2000), the Global Financial Crisis (2007–2009), and the COVID-19 pandemic (2020–2022). Results are shown for the full currency system as well as for selected subsamples, including the G3 currencies and the remaining currencies.

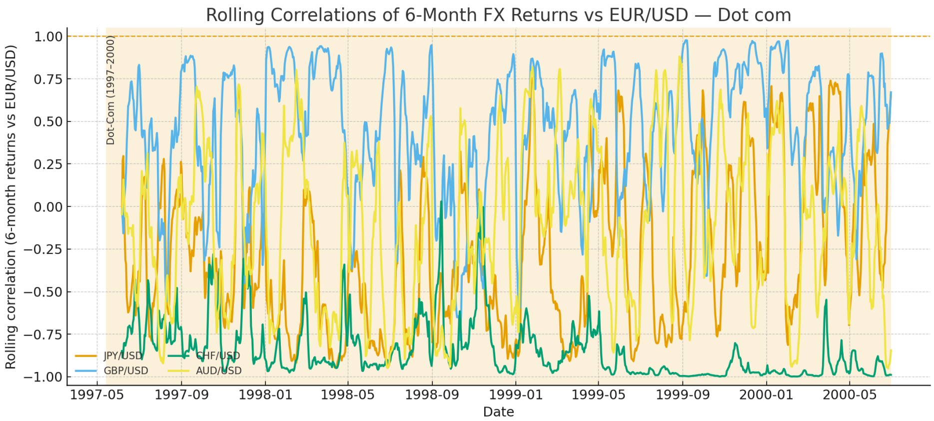

Figure 6: dot-com crash (1997–2000).

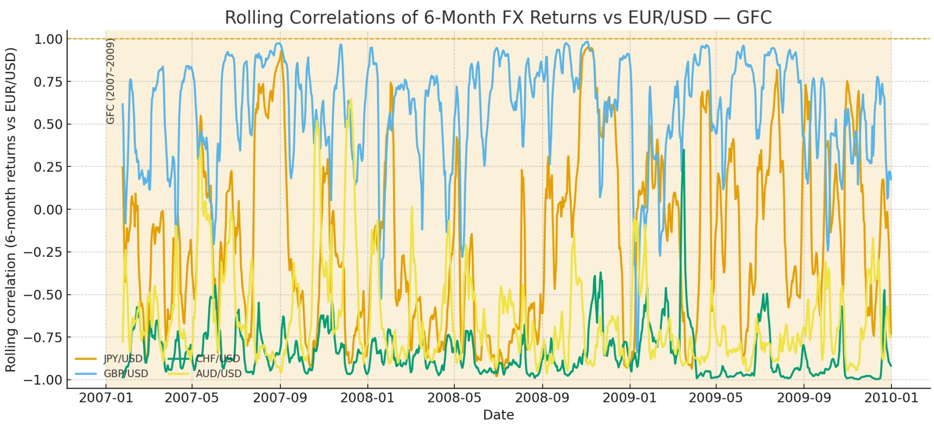

Figure 7: GFC (2007–2009).

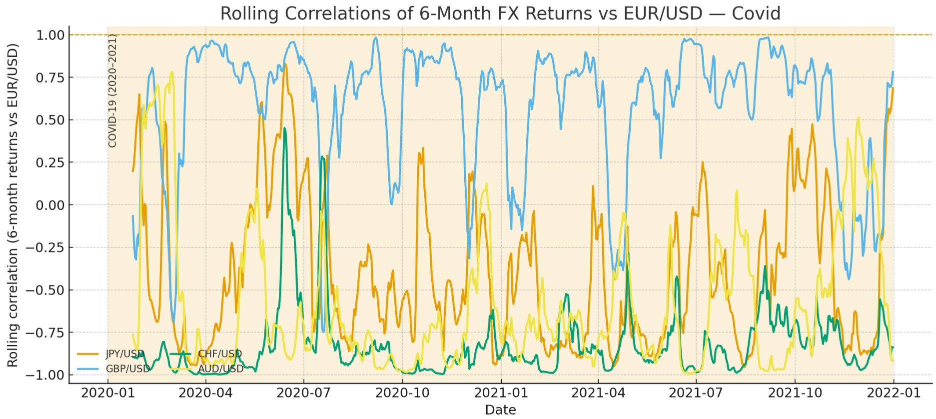

Figure 8: COVID-19 pandemic (2020–2022).

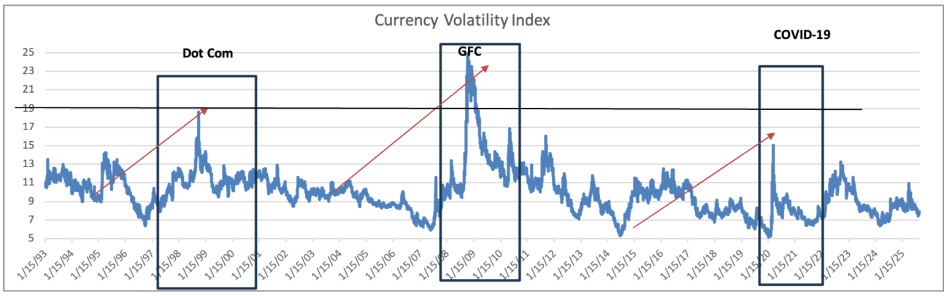

Figure 9: JP Morgan Global Currency Volatility Index with the dot-com crash, the Global Financial Crisis, and the COVID-19 pandemic windows highlighted.

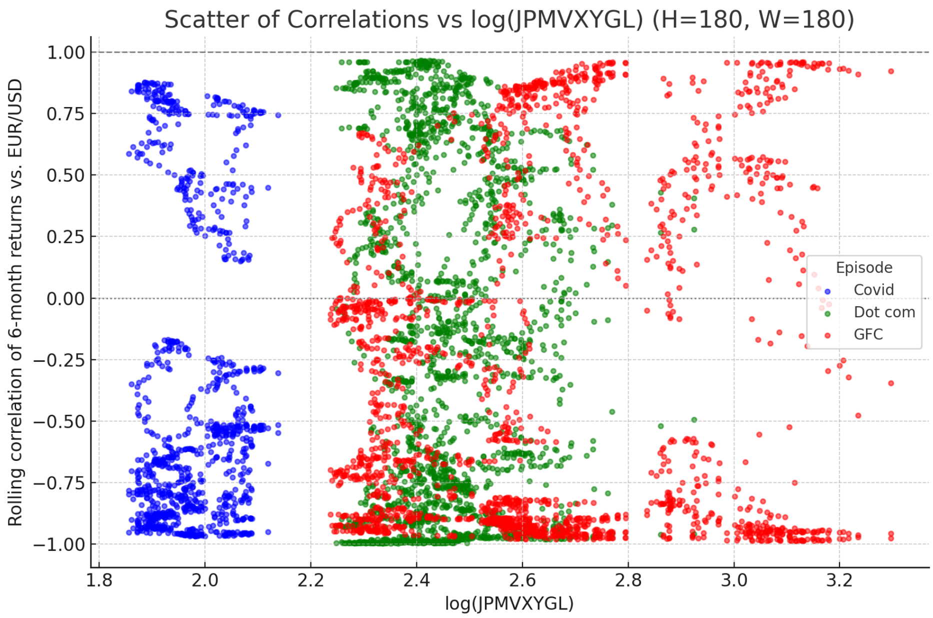

Figure 10: Scatter plot of rolling six-month correlations versus the log of the JPMorgan Global FX Volatility Index (JPMVXYGL). Each point represents a rolling six-month observation. Points are colored by crisis episode: red for the Global Financial Crisis (2007–2009), yellow for the dot-com crash (1997–2000), and blue for the COVID-19 pandemic (2020–2022).

Rank	Currency	Share of Global Turnover (%)	Notes
1	USD	89.2	Dominant global vehicle currency
2	EUR	28.9	Major reserve currency
3	JPY	16.8	Global safe haven
4	GBP	10.2	Liquid reserve-adjacent currency
5	CHF	6.4	Safe-haven currency
6	AUD	6.1	Commodity-linked global currency

Table 1: Foreign Exchange Market Turnover by Currency - Daily averages, April 2025, BIS Triennial Survey. **Notes:** This table reports the share of global foreign exchange market turnover by currency, measured as the percentage share of total average daily trading volume. Turnover shares sum to more than 100% because each foreign exchange transaction involves two currencies. The data are based on average daily turnover from the April 2025 Bank for International Settlements (BIS) Triennial Central Bank Survey (Bank for International Settlements, 2025) and include spot, forward, and swap transactions aggregated across reporting dealers.

	ADF Test		PP Test	
Variable	**t-Stat**	**p-val**	**Adj. t-Stat**	**p-val**
EUR/USD	-0.938263	0.7762	-0.903896	0.7873
Δ(EUR/USD)	-7.386294***	0.0000	-34.05348***	0.0000
JPY/USD	-1.141525	0.7011	-1.204249	0.6746
Δ(JPY/USD)	-15.27350***	0.0000	-31.62116***	0.0000
GBP/USD	-1.999517	0.2872	-1.893400	0.3120
Δ(GBP/USD)	-8.889021***	0.0000	-7.234940***	0.0000
CHF/USD	-1.385773	0.5906	-1.330779	0.6170
Δ(CHF/USD)	-23.15413***	0.0000	-34.46922***	0.0000
AUD/USD	-2.542213	0.1058	-2.477434	0.1213
Δ(AUD/USD)	-18.43708***	0.0000	-36.28420***	0.0000

Table 2: Unit root tests (dot-com crash). **Notes:** This table reports Augmented Dickey–Fuller (ADF) and Phillips–Perron (PP) unit root test statistics for major USD exchange rates during the dot-com crash. Tests are conducted on exchange rate levels and first differences. The null hypothesis for both tests is the presence of a unit root. Daily data are used over sample period spanning 1997–2000. Statistical significance is denoted by ***, **, and * for the 1%, 5%, and 10% levels, respectively.

Variable	ADF Test		PP Test	
	t-Stat	p-val	Adj. t-Stat	p-val
EUR/USD	-1.756445	0.4025	-1.821258	0.3703
Δ(EUR/USD)	-32.02499***	0.0000	-32.02657***	0.0000
JPY/USD	-1.061106	0.7328	-1.081701	0.7249
Δ(JPY/USD)	-18.10110***	0.0000	-36.10449***	0.0000
GBP/USD	-0.989019	0.7590	-0.938537	0.7761
Δ(GBP/USD)	-29.92993***	0.0000	-29.84658***	0.0000
CHF/USD	-1.614304	0.4749	-1.574312	0.4954
Δ(CHF/USD)	-33.79487***	0.0000	-33.82532***	0.0000
AUD/USD	-1.248617	0.6550	-1.317173	0.6235
Δ(AUD/USD)	-35.55946***	0.0000	-35.50432***	0.0000

Table 3: Unit root tests (Global Financial Crisis). **Notes:** This table reports Augmented Dickey–Fuller (ADF) and Phillips–Perron (PP) unit root test statistics for major USD exchange rates during the Global Financial Crisis. Tests are conducted on exchange rate levels and first differences. The null hypothesis for both tests is the presence of a unit root. Daily data are used over the sample period spanning 2007–2009. Statistical significance is denoted by ***, **, and * for the 1%, 5%, and 10% levels, respectively.

Variable	ADF Test		PP Test	
	t-Stat	p-val	Adj. t-Stat	p-val
EUR/USD	-1.519609	0.5233	-1.680846	0.4406
Δ(EUR/USD)	-10.66765***	0.0000	-24.81385***	0.0000
JPY/USD	-0.750186	0.8316	-1.142213	0.7006
Δ(JPY/USD)	-10.81124***	0.0000	-28.37586***	0.0000
GBP/USD	-1.463991	0.5515	-1.750368	0.4054
Δ(GBP/USD)	-10.32505***	0.0000	-22.62324***	0.0000
CHF/USD	-1.988275	0.2921	-2.173474	0.2164
Δ(CHF/USD)	-10.40580***	0.0000	-25.50979***	0.0000
AUD/USD	-1.676315	0.4429	-1.578366	0.4932
Δ(AUD/USD)	-15.93793***	0.0000	-23.58878***	0.0000

Table 4: Unit root tests (COVID-19 pandemic). **Notes:** This table reports Augmented Dickey–Fuller (ADF) and Phillips–Perron (PP) unit root test statistics for major USD exchange rates during the COVID-19 pandemic. Tests are conducted on exchange rate levels and first differences. The null hypothesis for both tests is the presence of a unit root. Daily data are used over the sample period spanning 2020–2022. Statistical significance is denoted by ***, **, and * for the 1%, 5%, and 10% levels, respectively.

Hypothesized No. of CE(s)	Eigenvalue	Max–Eigen Statistic	Critical Value (5%)	Prob.
None	0.020742	49.23901	76.97277	0.8738
At most 1	0.013552	30.12296	54.07904	0.9070
At most 2	0.009714	17.67890	35.19275	0.8560
At most 3	0.007319	8.77615	20.26184	0.7570
At most 4	0.002275	2.077047	9.164546	0.7623

Table 5: Unrestricted cointegration rank test (Trace) — dot-com crash. **Notes:** This table reports Johansen trace test statistics for assessing the number of cointegrating relationships among major USD exchange rates during the dot-com crash. For each hypothesized cointegration rank, the null hypothesis is that the number of cointegrating vectors is less than or equal to the stated rank. The table presents trace statistics, corresponding 5% critical values, and associated p-values. The analysis is based on daily spot exchange rate data covering the period 1997–2000. P-values are reported in the Prob. column.

Hypothesized No. of CE(s)	**Eigenvalue**	**Max–Eigen Statistic**	**Critical Value (5%)**	**Prob.**
None	0.020742	19.11606	34.80587	0.8639
At most 1	0.013552	12.44406	28.58808	0.9484
At most 2	0.009714	8.902748	22.29962	0.9096
At most 3	0.007319	6.699102	15.89210	0.7061
At most 4	0.002275	2.077047	9.164546	0.7623

Table 6: Unrestricted cointegration rank test (Maximum–eigenvalue) — dot-com crash. **Notes:** This table reports Johansen maximum-eigenvalue test statistics for assessing the number of cointegrating relationships among major USD exchange rates during the dot-com crash. For each hypothesized cointegration rank, the null hypothesis is that the number of cointegrating vectors is equal to the stated rank, against the alternative of one additional cointegrating vector. The table presents maximum-eigenvalue statistics, corresponding 5% critical values, and associated p-values. The analysis is based on daily spot exchange rate data covering the period 1997–2000. P-values are reported in the Prob. column.

Hypothesized No. of CE(s)	**Eigenvalue**	**Trace Statistic**	**Critical Value (5%)**	**Prob.**
None	0.031362	72.16038**	69.81889	0.0321
At most 1	0.020190	37.36500	47.85613	0.3305
At most 2	0.010765	15.09216	29.79707	0.7741
At most 3	0.002095	3.272948	15.49471	0.9531
At most 4	0.000899	0.982479	3.841465	0.3216

Table 7: Unrestricted cointegration rank test (Trace) — Global Financial Crisis. **Notes:** This table reports Johansen trace test statistics for assessing the number of cointegrating relationships among major USD exchange rates during the Global Financial Crisis. For each hypothesized cointegration rank, the null hypothesis is that the number of cointegrating vectors is less than or equal to the stated rank. The table reports trace statistics, corresponding 5% critical values, and associated p-values. The analysis is based on daily spot exchange rate data covering the period 2007–2009. P-values are reported in the Prob. column.

Hypothesized No. of CE(s)	**Eigenvalue**	**Max–Eigen Statistic**	**Critical Value (5%)**	**Prob.**
None	0.031362	34.79538**	33.87687	0.0388
At most 1	0.020190	22.27284	27.58434	0.2067
At most 2	0.010765	11.81921	21.13162	0.5656
At most 3	0.002095	2.290468	14.26460	0.9825
At most 4	0.000899	0.982479	3.841465	0.3216

Table 8: Unrestricted cointegration rank test (Maximum–eigenvalue) — Global Financial Crisis. **Notes:** This table reports Johansen maximum–eigenvalue test statistics for assessing the number of cointegrating relationships among major USD exchange rates during the Global Financial Crisis. For each hypothesized cointegration rank, the null hypothesis is that the number of cointegrating vectors equals the stated rank against the alternative of one additional cointegrating relationship. The table reports maximum–eigenvalue statistics, corresponding 5% critical values, and associated p-values. The analysis is based on daily spot exchange rate data covering the period 2007–2009. P-values are reported in the Prob. column.

Hypothesized No. of CE(s)	Eigenvalue	Trace Statistic	Critical Value (5%)	Prob.
None	0.037174	58.57631	69.81889	0.2817
At most 1	0.017760	27.51236	47.85613	0.8342
At most 2	0.010978	12.81844	29.79707	0.8997
At most 3	0.004213	3.766664	15.49471	0.9214
At most 4	0.000371	0.304418	3.841465	0.5811

Table 9: Unrestricted cointegration rank test (Trace) — COVID-19 pandemic. **Notes:** This table reports Johansen trace test statistics for assessing the number of cointegrating relationships among major USD exchange rates during the COVID-19 pandemic. At each step, the null hypothesis is that the cointegrating rank is less than or equal to the stated number. The table reports trace statistics, corresponding 5% critical values, and associated p-values for each hypothesized rank. The analysis is based on daily spot exchange rate data covering the period 2020–2022. P-values are reported in the Prob. column.

Hypothesized No. of CE(s)	Eigenvalue	Max–Eigen Statistic	Critical Value (5%)	Prob.
None	0.037174	31.06395	33.87687	0.1045
At most 1	0.017760	14.69392	27.58434	0.7720
At most 2	0.010978	9.051779	21.13162	0.8282
At most 3	0.004213	3.462246	14.26460	0.9113
At most 4	0.000371	0.304418	3.841465	0.5811

Table 10: Unrestricted cointegration rank test (Maximum–eigenvalue) — COVID-19 pandemic. **Notes:** This table reports Johansen maximum-eigenvalue test statistics for assessing the number of cointegrating relationships among major USD exchange rates during the COVID-19 pandemic. At each step, the null hypothesis is that the cointegrating rank equals the stated number of cointegrating relations against the alternative of one additional relation. The table reports maximum-eigenvalue statistics, corresponding 5% critical values, and associated p-values for each hypothesized rank. The analysis is conducted using daily spot exchange rate data covering the period 2020–2022. P-values are reported in the Prob. column.

Variable	Coefficient	Std. Error	t-Statistic
EUR/USD	1.000000		
AUD/USD	-0.834022	0.20812	-4.00746
CHF/USD	-1.686385	0.24968	-6.75420
GBP/USD	-0.374197	0.22057	-1.69649
JPY/USD	0.008727	0.00224	3.90429
Constant	1.161624		

Table 11: Johansen cointegrating vector (Global Financial Crisis; $r = 1$), normalized on EUR/USD. **Notes:** This table reports the estimated cointegrating vector for major USD exchange rates during the Global Financial Crisis under the restriction that the cointegration rank equals $r = 1$. The vector is normalized on the EUR/USD exchange rate, imposing a unit coefficient on EUR/USD. Coefficients on the remaining exchange rates, therefore, measure their long-run equilibrium relationship relative to EUR/USD. The table reports coefficient estimates, standard errors, and corresponding t-statistics for the non-normalized exchange rates only. The estimation is based on daily spot exchange rate data covering the period 2007–2009.

	Dependent Variable				
	Δ**EUR/USD**	Δ**AUD/USD**	Δ**CHF/USD**	Δ**GBP/USD**	Δ**JPY/USD**
u_{t-1}	0.030***	−0.016	0.050***	0.020***	0.689
	(0.007)	(0.012)	(0.011)	(0.006)	(1.061)
	[4.561]	[−1.304]	[4.632]	[3.355]	[0.656]
ΔEUR_{t-1}	0.189**	−0.216	0.245*	0.042	3.647
	(0.0916)	(0.1684)	(0.1473)	(0.0817)	(14.4895)
	[2.060]	[−1.284]	[1.666]	[0.510]	[0.252]
ΔEUR_{t-2}	−0.197**	0.295*	−0.194	−0.129	14.616
	(0.0905)	(0.1663)	(0.1455)	(0.0807)	(14.3113)
	[−2.174]	[1.771]	[−1.336]	[−1.599]	[1.021]
ΔAUD_{t-1}	0.013	−0.168***	0.006	0.031	−10.751**
	(0.0332)	(0.0611)	(0.0534)	(0.0296)	(5.2553)
	[0.389]	[−2.750]	[0.121]	[1.048]	[−2.046]
ΔAUD_{t-2}	0.044	−0.072	0.059	0.022	−8.853*
	(0.0332)	(0.0611)	(0.0534)	(0.0296)	(5.2543)
	[1.326]	[−1.185]	[1.106]	[0.736]	[−1.685]
ΔCHF_{t-1}	−0.077	−0.000	−0.108	−0.026	−6.383
	(0.0500)	(0.0920)	(0.0805)	(0.0446)	(7.9141)
	[−1.544]	[−0.002]	[−1.347]	[−0.578]	[−0.807]
ΔCHF_{t-2}	0.087*	−0.235**	0.077	0.064	−16.633**
	(0.0499)	(0.0917)	(0.0802)	(0.0445)	(7.8895)
	[1.746]	[−2.567]	[0.958]	[1.438]	[−2.108]
ΔGBP_{t-1}	−0.065	0.057	0.018	0.110**	11.921
	(0.0573)	(0.1054)	(0.0922)	(0.0511)	(9.0658)
	[−1.135]	[0.546]	[0.195]	[2.156]	[1.315]
ΔGBP_{t-2}	0.150***	−0.195*	0.211**	0.083*	−17.228*
	(0.0568)	(0.1045)	(0.0914)	(0.0507)	(8.9923)
	[2.641]	[−1.869]	[2.308]	[1.635]	[−1.916]
ΔJPY_{t-1}	−0.000033	0.000918	−0.000554	−0.000668**	−0.02286
	(0.00033)	(0.00061)	(0.00053)	(0.00029)	(0.05208)
	[−0.101]	[1.517]	[−1.047]	[−2.276]	[−0.439]
ΔJPY_{t-2}	−0.000544*	0.001989***	−0.000329	−0.000608**	0.1652***
	(0.00033)	(0.00061)	(0.00053)	(0.00030)	(0.05257)
	[−1.637]	[3.257]	[−0.615]	[−2.053]	[3.142]
Constant	−0.0000456	−0.0000129	−0.000225	0.000103	−0.03258
	(0.00019)	(0.00035)	(0.00030)	(0.00017)	(0.02995)
	[−0.241]	[−0.037]	[−0.740]	[0.610]	[−1.088]
R^2	0.0519	0.0312	0.0584	0.0511	0.0536

Table 12: Vector error correction model short-run dynamics (Global Financial Crisis; dependent variables in columns). **Notes:** This table reports coefficient estimates from the vector error correction model estimated for major USD exchange rates during the Global Financial Crisis; standard errors are reported in parentheses, and corresponding t-statistics are reported in brackets. The variable u_{t-1} denotes the lagged error-correction term constructed from the cointegrating vector reported in Table 11. The estimation is based on daily spot exchange rate data covering the period 2007–2009. Statistical significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively.

	EUR/USD	JPY/USD	GBP/USD	CHF/USD	AUD/USD
$H = 1$ day ahead	-0.20	0.02	-0.09	-0.25	0.01
$H = 7$ days ahead	-0.19	0.05	-0.10	-0.26	0.02
$H = 30$ days ahead	-0.25	-0.03	-0.18	-0.25	-0.03
$H = 90$ days ahead	-0.26	-0.04	-0.17	-0.22	0.02
$H = 180$ days ahead	-0.13	0.02	-0.11	-0.29	0.13
$H = 365$ days ahead	0.12	0.34	0.23	-0.56	0.31

Table 13: dot-com crash: Average Correlation of H-day-ahead foreign exchange returns (in USD). **Notes:** This table reports average correlations between H-day-ahead foreign exchange returns, measured in USD, and the benchmark return used in the forecasting exercise during the dot-com crash. Each entry corresponds to the mean correlation computed at the specified forecast horizon H. Correlations are calculated using daily data and summarize the degree of comovement between currency returns and the benchmark over different horizons. The sample covers the dot-com crash from 1997 to 2000.

	EUR/USD	JPY/USD	GBP/USD	CHF/USD	AUD/USD
$H = 1$ day ahead	-0.24	0.00	-0.08	-0.13	-0.30
$H = 7$ days ahead	-0.25	0.01	-0.07	-0.14	-0.30
$H = 30$ days ahead	-0.28	0.05	-0.10	-0.11	-0.28
$H = 90$ days ahead	-0.24	0.09	-0.06	-0.15	-0.34
$H = 180$ days ahead	-0.20	0.07	-0.08	-0.22	-0.36
$H = 365$ days ahead	-0.26	0.02	-0.20	-0.20	-0.29

Table 14: GFC: Average Correlation of H-day-ahead foreign exchange returns (in USD). **Notes:** This table reports average correlations between H-day-ahead foreign exchange returns, measured in USD, and the benchmark return used in the forecasting exercise during the Global Financial Crisis. Each entry corresponds to the mean correlation computed at the specified forecast horizon H. Correlations are calculated using daily data and summarize the degree of comovement between currency returns and the benchmark across different horizons. The sample period spans the Global Financial Crisis from 2007 to 2009.

	EUR/USD	JPY/USD	GBP/USD	CHF/USD	AUD/USD
$H = 1$ day ahead	-0.34	-0.03	-0.25	-0.08	-0.16
$H = 7$ days ahead	-0.36	-0.01	-0.28	-0.07	-0.15
$H = 30$ days ahead	-0.33	0.02	-0.20	-0.07	-0.22
$H = 90$ days ahead	-0.37	0.05	-0.16	-0.02	-0.19
$H = 180$ days ahead	-0.48	0.11	-0.20	0.08	-0.09
$H = 365$ days ahead	-0.48	0.07	-0.26	0.04	-0.07

Table 15: COVID-19 pandemic: Average Correlation of H-day-ahead FX returns (in USD). **Notes:** This table reports average correlations between H-day-ahead foreign exchange returns, measured in USD, and the benchmark return used in the forecasting exercise during the COVID-19 pandemic. Each entry corresponds to the mean correlation computed at the specified forecast horizon H. Correlations are calculated using daily data and summarize the degree of comovement between currency returns and the benchmark across different horizons. The sample period spans the COVID-19 pandemic from 2020 to 2022.